User Guide

Autodesk Inventor integration
for
Aras Innovator

Valid for product version: 4.7.0

Published: 27.02.2026 | Build: 1285 | Revision: 4bb116954

XPLM Solution GmbH

Altmarkt-Galerie

Altmarkt 25

01067 Dresden

Office: +49 351 82658-0

Fax: +49 351 82658-88

Web: www.xplm.com

Legal information

Non-disclosure

The information in this document is to be treated confidentially.

Licensing

XPLM Solution GmbH (XPLM) does not issue licenses for the software components to be integrated. These licenses must be obtained from the respective software manufacturer. To license the integration software from XPLM, a software usage and maintenance contract must be concluded.

Support

In the event of problems with the integration software, access the XPLM Support Portal at https://support.xplm.com and file a ticket. If you don't have an account yet, contact support@xplm.com and request one.

Feedback

If you discover any errors or inconsistencies in this documentation, write to documentation@xplm.com and let us know what we can improve.

Handling personal data

Against the background of the EU General Data Protection Regulation (GDPR), XPLM informs about the processing of personal data in its contracts and privacy policies. The extent to which XPLM comes into contact with personal data from the customer environment while providing services depends on the customer's IT systems and the data transmitted to XPLM. XPLM is generally not interested in gaining access to personal data from the customer environment. If this cannot be avoided, the customer is responsible for such personal data and must enter into a GDPR-compliant data processing agreement with XPLM.

Information exchange: The integration software exchanges information between the systems involved in the form of attributes. What is contained in these attributes is left to the customer's definition. For ECAD integrations, the exchanged attribute information is saved in the file `metadata.xml` in the configured project directory or directly in the project files.
For other integrations, the exchanged attribute information is saved in files or objects of the source system.

Log files: To record activities of integration functions, the integration software stores log files on the computer on which it is installed. These log files can also contain information from Windows environment variables, for example the user and computer name. For ECAD integrations, log files are stored in the directory `%TEMP%\integrate` by default. This directory can be deleted, but new log files will be created when the integration software is used again.
For other integrations, logging is deactivated by default. Activation, protocol level and storage location must be set up in advance. Directories with log files can be deleted, but new log files will be created when the integration software is used again.

Support tickets: When submitting a support ticket in the event of problems with the integration software, information such as first and last name, email address, etc. can be transmitted to XPLM support. For more information on this topic, see *Legal Notices* > *Privacy Policy* in the support portal.

Other policies: XPLM also refers to the privacy policies of the software manufacturers involved for the handling of personal data in their systems.

Table of Contents

Glossary

Application Programming Interface (API)

Defines a set of routines, communication protocols and tools for building software. In general, they are clearly defined methods for communication between different components.

Aras Markup Language (AML)

Defines the backbone language of Aras Innovator. Almost every action a user performs in the client sends an AML request to the Aras Innovator server to process the business logic.

Bill of Materials (BOM)

Defines a list of assemblies, sub-assemblies, parts and their quantities needed to produce a final product.

BOM position

Defines a position in the BOM with unique identification, name, quantity and other characteristics.

Component Object Model (COM)

Defines a binary-interface standard for software components introduced by Microsoft.

Connector

Defines a central interface component of each XPLM integration. The integration uses connectors for each participating application to exchange data via their API.

Datamodel

Defines objects and their relationships in a PLM system that are managed by the integration to store data from an authoring application.

Dynamic Link Library (DLL)

Defines a file with a library of functions and other information that can be accessed by a Windows program.

Generation

Defines an incremental counter of each object modification in Innovator on check-in.

Payload

Defines the data contained within an API request. The description is borrowed from the transportation industry, where a truck carries its cargo (its payload) to a location. The truck, as with the API request, is always the same, but the payload changes with each request.

Product Lifecycle Management (PLM)

Defines systems and processes for managing data during the development of a product from creation through manufacturing to maintenance and disposal.

Revision

Defines a released object state in Innovator that cannot be modified.

Script engine

Defines the central component in each integration. It contains the integration logic for processing and forwarding the information and data coming from the connectors.

User Interface (UI)

Defines a (usually) graphical interface through which a user interacts with the computer.

x86/x64

Defines the processor architecture in a computer and thus also the performance of applications. x86 corresponds to 32-bit and x64 corresponds to 64-bit.

1 Introduction

1.1 About this document

You are reading the User Guide of the Autodesk Inventor integration for Aras Innovator.

Purpose

This document shows how to use the integration.

Target audience

This document is intended for engineers in the company, who use the integration on a daily basis. System administrators should also read this document to familiarize themselves with the user interface and integration functions.

How to read this document

This document is structured chronologically and you should read it in the order of the chapters described. If you skip chapters, you will miss important information.
Where appropriate, cross-references to other chapters are listed. To quickly return to where you came from after clicking such a link, click the **Back** button in your PDF viewer. Try it right now with this link!

Notes used

This note highlights additional information about the current content.

This note highlights important instructions.

1.2 Naming conventions

In addition to the terms already listed in the glossary, the following naming conventions are used in this document. Familiarize yourself with these names. You will find them throughout the document.

Form Used	Generally refers to
Innovator	Aras Innovator
Inventor	Autodesk Inventor
Part	Part in Innovator
Attribute	Object property in Innovator
Library Part	Standard Part

1.3 What's new

Check out what's new in this release! This section provides a summary of the latest documentation updates based on new features and improvements to the integration. Refer to the official release notes for detailed release information.

Version 4.7.0

One-click Load, Insert and Replace in Innovator
A **One-Click** > **Load/Insert CAD/Replace** button has been introduced in Innovator Web Client to streamline CAD integration workflows. This feature enables users to quickly load, insert or replace CAD components directly from Innovator

Highlights

- One-Click load from Innovator (p. 47) - Quickly load selected CAD Document to Inventor.

Optimized Synchronize Part Library
The **Synchronize Part Library** command now downloads only new or modified Standard Parts instead of all parts in Innovator.

Highlights:

- Only parts with a `caxtimestamp` greater than the last synchronization timestamp are downloaded.
- Stores last sync timestamp in `StandardPartsLastSync.txt.`
- Displays a confirmation dialog if many parts need to be downloaded.

Learn more: Innovator menu (p. 10)

Save older generations as newest
You can now restore older generations of Inventor objects and create a new generation directly from the *Save dialog*.

Highlights:

- New context menu command, **Restore generation** in *Save dialog*.
- Supports multi-select for bulk operations.
- Includes permission checks and confirmation dialogs.
- Automatically updates CLB files and marks objects for saving.

Learn more: Restoring older generation as newest (p. 41)

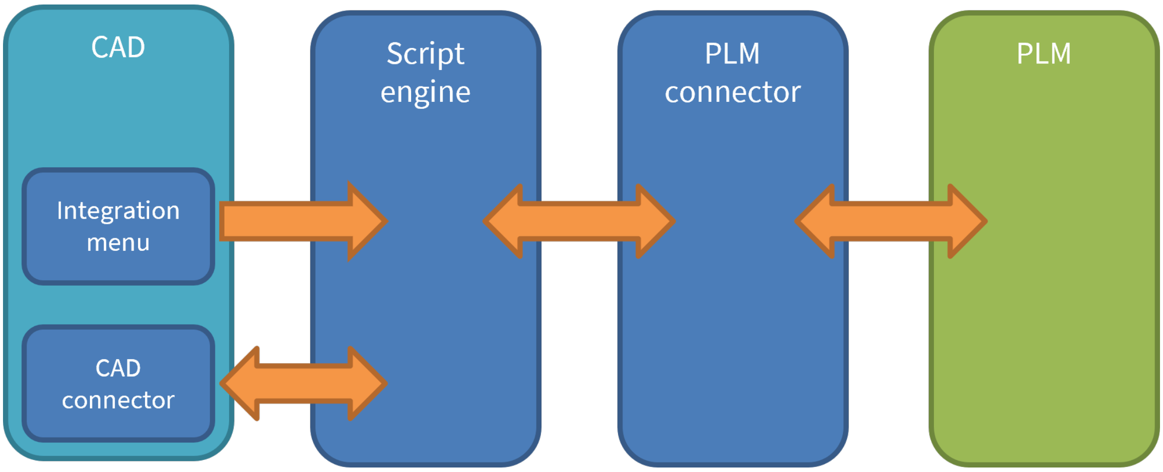
1.4 Integration overview

The integration is the interface between Inventor and Innovator and is designed for seamless and efficient data exchange between both applications.

The integration focuses on a Inventor-centric workflow, enabling users to interact with Innovator directly within Inventor. The main integration functions include saving, loading, and performing object operations. When saving CAD data to Innovator, corresponding objects and structures are created in Innovator and linked to the CAD data. The original CAD files are attached to these objects, allowing for centralized data management within the company.

To support concurrent engineering and avoid access conflicts, the integration utilizes the object reservation capabilities (checkin/checkout) from Innovator.

The integration functionalities are accessible through additional toolbars or extended menus in Inventor.



1.5 Supported file types

The Innovator - Inventor integration supports the following file types:

File type	File extension
Part	.ipt
Assembly	.iam
Drawing	.idw .dwg

2 Functional description

2.1 Innovator menu

After installing the integration, the integration ribbon menu appears on the Inventor user interface. Additional steps may be required to make the integration menu visible. See Installation and Administration Guide for more information.

The integration menu contains the following commands:

Connection

	Connection Parameters	Opens a dialog to select a pre-defined environment with the URL to Innovator.
	Connect	Opens the Innovator login dialog.
	Disconnect	Terminates the connection to Innovator
	Start Client	Opens the Innovator web client.

New

	New Part	Creates a new part, saves it to Innovator and opens the file in Inventor.
	New Assembly	Creates a new assembly, saves it to Innovator and opens the file in Inventor.
	New Drawing	Creates a new drawing, saves it to Innovator and opens the file in Inventor.

Save

	Save	Saves files from the current active Inventor model and all its components into Innovator, with a dialog that allows pre-selection of files to save.
	Save Session	Saves all opened and modified files from the current Inventor session to Innovator. The difference between **Save Session** and **Save** is that all objects (not only the active ones) are saved to Innovator. All objects are closed after this process.
	Save (1st Level)	Saves the top level and first level elements to Innovator. The structure must be known to Innovator already for this function to work.
	Copy	Clones the current Innovator-known active Inventor model with a dialog that allows pre-selection of objects to cloned.

Load

	Load from Document	Launches the *Load* dialog that allows you to search for Innovator CAD documents to download to the active workspace.

	Load from Part	Launches the *Load* dialog that allows you to search for Innovator part objects to download to the active workspace
	Load from Client	Launches the *Load* dialog that displays CAD document added to the Load queue from Innovator.
	Load related Drawings	Performs a query in Innovator of the current Inventor model and filters related drawings. The resulting drawings are added to the **Load Preview** from which the drawings can be selected and loaded to the local workspace. The latest generation of drawings is always retrieved, even across revision boundaries. For example Gen 3 of the drawing is retrieved for Gen 1 of the model.
	Load related Assemblies	Performs a query in Innovator of the current Inventor model and filters related assemblies. The resulting assemblies are added to the **Load Preview** from which the assemblies can be selected and loaded to the local workspace.
	Update Workspace	Updates files in the local workspace with the latest generation from Innovator with a dialog that enables you to select the files to update.

Access Control

	Claim Claim (Structure)	Sets reservations on either a single document or an entire document structure in Innovator for the current user.
	Unclaim Unclaim (Structure)	Removes reservations from a single document or an entire document structure only if the current user has reserved the documents. Reservations made by other users cannot be removed using these functions.

Property Update

	Current	Sets custom file property values in Inventor based on values from Innovator. Only properties for the current Inventor model are updated.
	Structure	Sets custom file property values in Inventor based on values from Innovator. Properties of the current Inventor model and all its child components are updated.

Material Management

	Create Part	Creates or updates part(s) in Innovator for the current Inventor model.
	Create Part (1st Level)	Creates or updates part(s) in Innovator for the first level elements of current structure.

	Create Part and BOM	Creates or updates BOM in Innovator for the current Inventor model. If no Parts are attached to the corresponding CAD Documents, they are created and assigned accordingly. Make sure related documents are claimed in Inventor before creating parts.
	Create Part and BOM (1st Level)	Creates or updates BOM in Innovator of the first level elements of the current structure.

Workspace

	Show Workspace	Opens the **Workspace Browser** which displays Innovator related information of the current Inventor model and all its components. The Workspace Manager's main task is to facilitate file management.
	Select Workspace	Enables you to choose any local directory as the current local working directory. It launches a directory selector window from which you can browse the available directories and select the desired folder.
	Remove Local Files	Deletes files from the local workspace with a confirmation dialog. If files are deleted, the corresponding .CLB files are also removed.

Tools

	Display Document	Launches the Innovator Web Client and displays the CAD Document of the current Inventor file.
	Display Part	Displays the part object linked to the currently opened CAD model in Innovator.
	Assign Document	Assigns a desired Innovator CAD Document to the current Inventor model. This function should only be used for empty or newly created CAD Documents. This function launches the *Load dialog* which allows you to search for the desired CAD Document to assign.
	Open Discussion Panel	Launches the **Discussion Panel** for the current Inventor model. The file must be known in Innovator. This is an easy-to-use social collaboration method using Aras Innovator Visual Collaboration discussions directly inside Inventor. You can view comments made relative to the active document and post further comments or replies. Refer to the corresponding Aras Innovator documentation for more information.

	Synchronize Part Library	Downloads and updates the standard parts in the defined *standard part check out* local directory with the corresponding part list from the Innovator database. This function downloads all Inventor objects marked as `is_standard` in Innovator when executed for the first time, otherwise only new or modified objects are downloaded (since the last synch) on subsequent runs. See Installation and Administration Guide for more information.
	Info	Provides basic information about the Inventor integration in use.

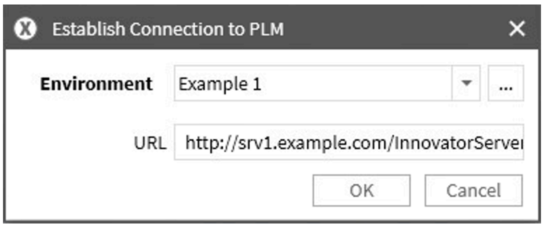
2.2 Login dialogs

Login to Innovator is now simplified. The URL to Innovator is read from an own dialog, whereas the actual connection is handled by the Aras login dialog.

Connection parameters

The function **Connection Parameters** shows the dialog **Establish Connection to PLM**. In this dialog, you can select a pre-defined environment with a URL to Innovator.

Figure 1: Establish Connection to PLM



- **Environment**: Shows a list of available environments.
- **[...]**: Shows ways to modify environments.
  - **Copy current data set**: Creates a copy of the current environment.
  - **Delete current data set**: Deletes the current environment.
  - **Reset data set**: Re-reads environments from file `XPlmArasLogonParameter.xml` in directory `%xPlmRootDir%\xml` and applies this information to the same file in the `%TEMP%` directory.
- **OK**: Confirms environment selection.
- **Cancel**: Closes dialog without further actions.

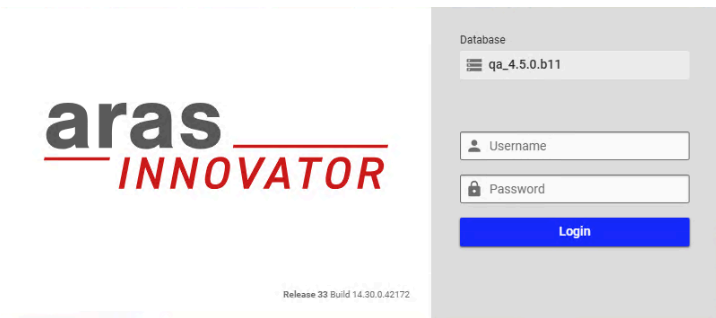
When executing this function for the first time, default environment definitions are read from the file `XPlmArasLogonParameter.xml` in the main configuration directory `%xPlmRootDir%\xml`. After reading, a second file with the same name is created in directory `%TEMP%\xplm`.

You can now modify connection parameters conveniently in this dialog, for example create new environments, delete existing or modify URLs. All changes are written to the file in the `%TEMP%` directory. The defaults in `%xPlmRootDir%\xml` are not affected and can be used to reset the connection parameters.

Aras login dialog

The selected environment is used in the function **Connect** to show the Aras login dialog. All further authentication happens in this dialog.

Figure 2: Aras login dialog

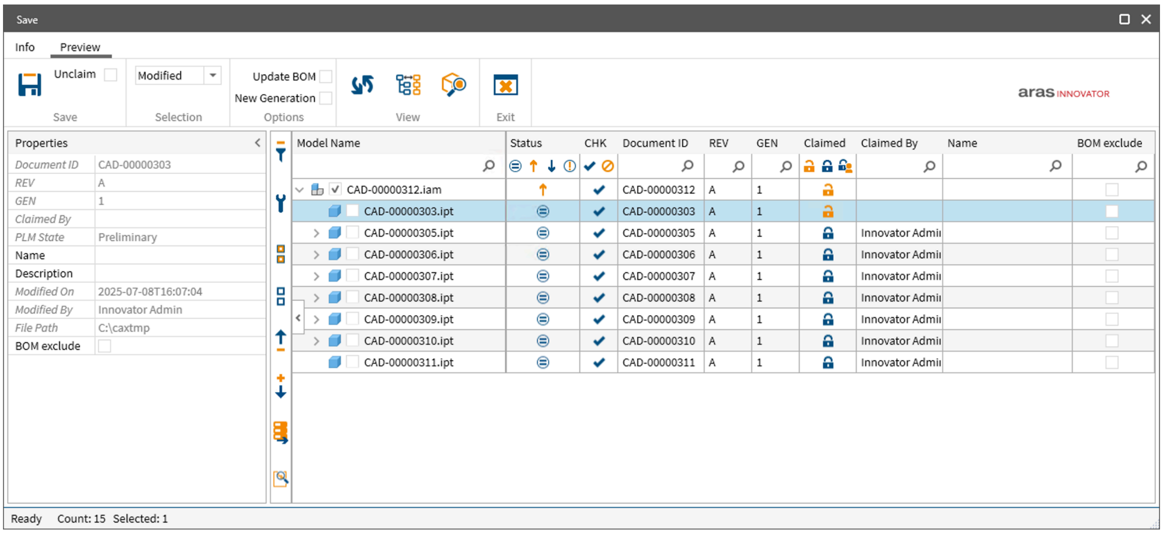


When you close Inventor, the integration also disconnects from Innovator and all processes are stopped.

2.3 Save dialog

In this dialog you can save Inventor data to Innovator and review related information.

The *Save dialog* is displayed by any of the **Save** commands within the integration menu.



The menu ribbon of the *Save dialog* has two tabs;

- Info tab (p. 28)
- Preview Tab (p. 15)

Sections in the *Save dialog* (from left to right):

- Properties section (Properties section/metadata table (p. 29))
- Side bar (p. 28) (collapsible)
- Model section (Model section (p. 16) & Properties section/metadata table (p. 29))

The *Save dialog* offers additional functionalities in the Model section through;

- Context menu (p. 32)
- Filter options (p. 32)

The bottom of the *Save dialog* has a **Status line** which displays the number of available and selected files in the model window

The appearance of the *Save* dialog can be changed by system administrators or users. See Installation and Administration Guide for more information.

2.3.1 Preview Tab

This section contains the main functions for saving model data in Innovator.

Function	Description
Save	■ - Starts the save operation to Innovator. ■ **Unclaim check box** - When checked, this function automatically releases reservation of the CAD Documents after a successful save to Innovator. This function is ignored on Initial save of a new document to Innovator. The document remains reserved, even if the check box is selected.

Function	Description
Selection	Selects/deselects the check box of line items in the model window based on their status or in general. Possible options in the drop-down menu include: ■ **Nothing**: Deselects everything ■ **Locked**: Selects only locked items ■ **All**: Selects everything ■ **Modified**: Selects only locally modified items Only line items with a selected check box are saved in Innovator.
Options Toggle buttons	■ **Update BOM** - If enabled, creates or updates BOM in Innovator for the selected line items when saving. If no Parts are attached to the corresponding CAD Documents, they are created and assigned accordingly. ■ **New generation** - If enabled, a new generation is created for each selected line item when saving.
View	■ **Refresh** - If model data is already saved, updates the metadata for all line items from Innovator. ■ **Structure/List View** - Toggles between structure and list view (flat list) in the model window. ■ **Show Occurrences** - Shows or hides multiple displayed objects level by level.
Exit	Closes the dialog without further actions.

2.3.2 Model section

This section shows the model data with its metadata in tabular form. When the save dialog is started, checks are performed on the model, status and possible actions are indicated for each file/object.

The model data is displayed in structure or list view.

Model section includes

- Metadata table with filter options
  - Possible columns are described here Properties section/metadata table (p. 29).
- Row items

Metadata table

Possible columns are described here Properties section/metadata table (p. 29).

Row items of metadata table

Model data is shown per row item in different columns. These selection methods are available:

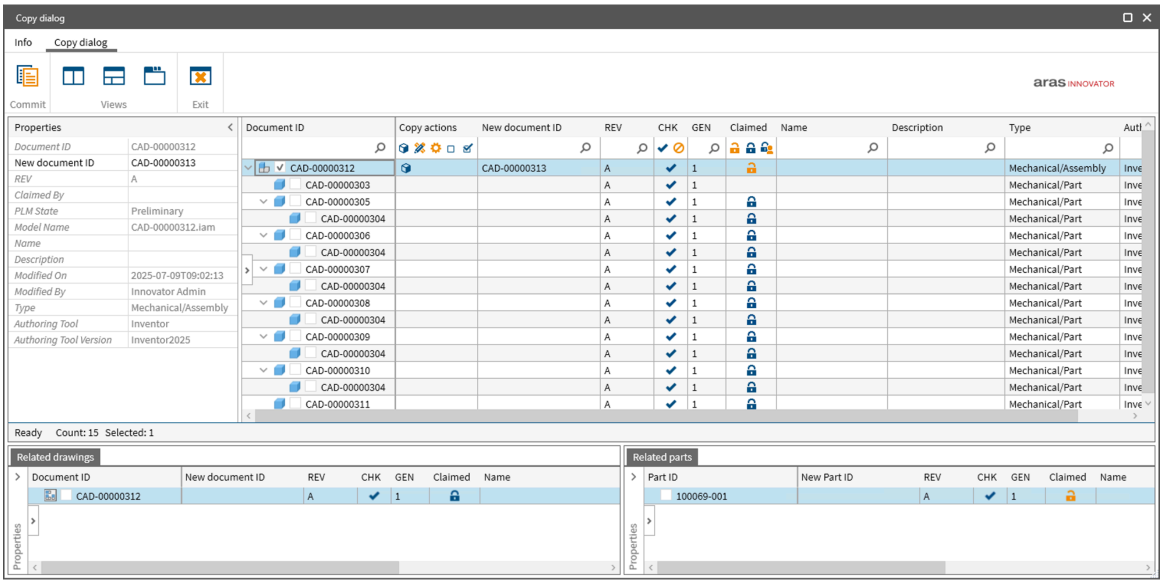
- A selected check box in front of a row item will mark it for further processes in Innovator.
- A selected row item allows for context menu functions. You can either select one row item or multi-select several items by either **CTRL**+left-click, or a sequence with **SHIFT**+left-click.

2.4 Copy dialog

This dialog allows you to copy documents, drawings and parts in Innovator. It also offers the possibility to edit properties of the newly created document.

The *Copy* dialog is displayed by the **Copy** command from:

- the integration menu
- the context menu of the *Load dialog*.



The *Copy dialog* has the following main sections:

- Menu Ribbon (p. 17)
- Model section (p. 18)

The *Copy dialog* offers additional functionality in the Model section through the;

- Filter options (p. 32)
- Context menu (p. 32)

2.4.1 Menu Ribbon

The ribbon contains the command buttons organized within a set of tabs. On each tab, the related buttons are grouped.

The menu ribbon of the *Copy dialog* has two tabs:

- Info tab (p. 28)
- Copy dialog tab

Copy dialog tab

This section contains the main functions for copying model data in Innovator and options to configure the layout of the *Copy dialog*.

Function	Description
Commit 	Starts the copy operation in Innovator and adds the property values set in the Copy dialog to the newly created copy.

<table>
<tr><th>Function</th><th>Description</th></tr>
<tr><td>Views</td><td>Contains options to customize and organize the different panels of the Copy dialog.

There are 3 main views:
Multi view - All the panels are displayed in grid like format with the Documents panel occupying the upper half of the main window. The Related drawings and Related parts panels are displayed in the lower half of the main window.
Mixed view - the Documents panel is docked at the top and the Related drawings and Related parts panels are tabbed in the lower half of the main window.
Tab view - All the panels are organized as individual tabs within the main window. You can switch between tabs to access the different panels.</td></tr>
<tr><td>Exit</td><td>Closes the dialog without further actions.</td></tr>
</table>

2.4.2 Model section

This section shows the model data with its metadata in tabular form.

The model window contains 3 main panels:

- Documents panel
- Related drawings panel
- Related parts panel

Each of the panels has the following sections:

- **Properties section** - Same behavior and functionality as the Properties section of the *Save dialog* except it has an additional property, *New document number* or *New part number* in case of the *Related parts panel* , in which you can manually enter the object ID of the newly created copy in Innovator.
- **Side bar** (collapsible) - Same behavior and functionality as the Side bar (p. 28) section of the *Save dialog*.
- **Metadata table** - shows model data with its metadata in tabular form. The columns provide additional details of Innovator properties related to the model, while the row items show the components of the Inventor structure. When a component is selected in the *Documents* panel, related drawings and parts Innovator objects are automatically shown in the respective panels and can also be selected for copy function.

The *Documents* panel has additional section at the bottom of the panel; **Status line** which shows the number of available and selected files in the metadata table.

Metadata table

Possible columns are described here Properties section/metadata table (p. 29).

Row items of metadata table

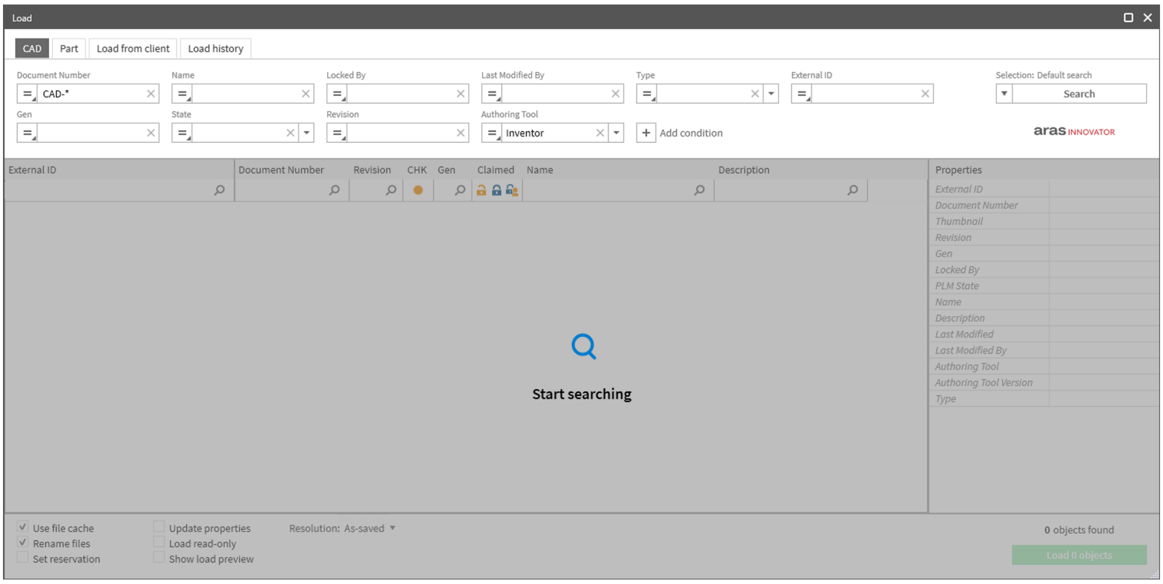
Model data is shown per row item in different columns. These selection methods are available:

- A selected check box in front of a row item will mark it for further processes in Innovator.

- A selected row item allows for context menu functions. You can either select one row item or multi-select several items by either **CTRL**+left-click, or a sequence with **SHIFT**+left-click.

2.5 Load dialog

In this dialog, you can review the Innovator metadata and choose model data to be be loaded to the local working directory.



Sections in the *Load dialog* include:

- Tabs (p. 19)
- Search toolbar (p. 20)
- Results section (p. 22)
- Properties section (Properties section/metadata table (p. 29))
- Load options (p. 22)

The *Load dialog* offers additional functionality in the Results section through the;

- Filter options (p. 32)
- Context menu (p. 32)

The appearance of the *Load dialog* can be changed by the system administrator; it does not necessarily have to appear like the one in the example above. See Installation and Administration Guide for more information.

2.5.1 Tabs

The *Load dialog* includes four tabs at the top.

Each tab corresponds to a different data category. They include;

- **CAD** - You can set search criteria for CAD Documents to be loaded based on properties of the CAD Documents in Innovator.
- **Part** - You can set search criteria for Part objects to be loaded based on properties of the Part objects in Innovator.
- **Load from client** - displays a list of files that have been added to the load queue by using the function **Load to CAD** in Innovator. You can select files to be loaded by either marking the check box or using the context menu function **Select**.
- **Load history** - displays a list of files that were last loaded via the *Load* dialog. You can select files to be reloaded by marking the check box in the **External ID** column or using the context menu function **Select**.

2.5.2 Search toolbar

Located below the tabs, this area offers multiple search conditions which can be removed or added as required.

The sections in the search bar include:

- Search criteria
- Search setups

Search criteria

In this section, you can define search criteria and start the search with the button **Search**.

If a search is executed without defining any search criteria, this will by default return all entries from the database until the defined search limit is reached.

Filter fields
The search criteria fields are only available in the following tabs of the *Load dialog*.

- **CAD**

- **Part**

The labels of the search fields are derived from Innovator. If a field has no label, its internal name is displayed instead.

You can add additional default search fields by clicking **Add condition**. You can also configure new search fields not in the default. See Installation and Administration Guide for more information.

To remove search fields, click within the search field to activate it, then on the `X` icon on the top right.

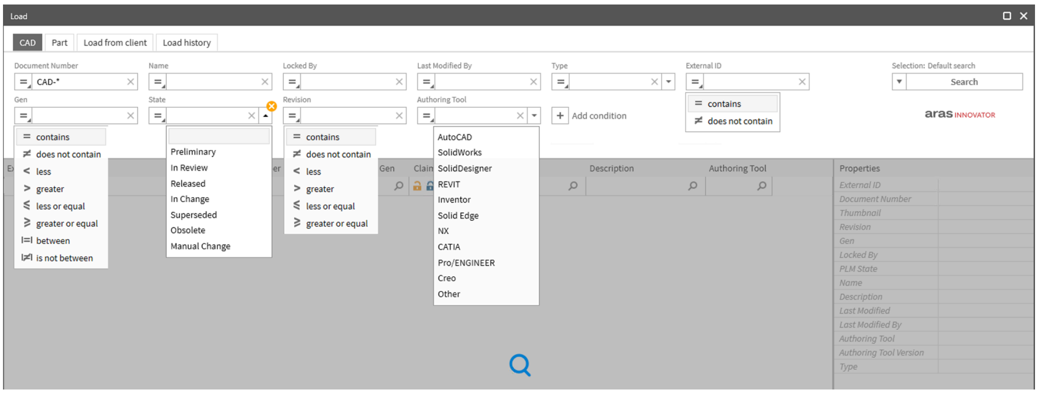
It is possible to use several search conditions to find required documents. Within the search field, the wild-card (*****) is supported.

Filter conditions
Each filter field has a drop-down menu of comparison operators on the left of the search field box, to narrow down the search as desired. The data type of the search field determines what operators are available. They include:

- **= equal** - returns results that match the values entered
- **≠ is not equal** - returns results that do not match the values entered
- **< less** - returns results with values less than the value entered.
- **> greater** - returns results with values greater than the value entered.
- **≤ less or equal** - returns results with values equal to or less than the value entered.
- **≥ greater or equal** - returns results with values equal to or greater than the value entered.
- **|=| between** - returns results within the range given
- **|≠| is not between** - returns results outside of the range given
- **Ø is null** - returns results that do not contain a value
- **is not null** - returns results that contain a value in the selected field

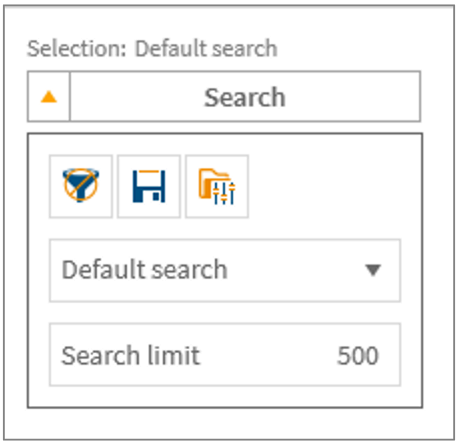
Some search fields contain valid and acceptable values already set. These are available on the right of the search field box in a drop down menu.

Example filter options;



Search setups

From the drop down menu on the button **Search**, you can define the search limit and save search setups.



The following functions are available within the Search drop down menu;

Function	Description
Clear Filter	Removes entered values in the search fields.
Save Search	Creates and saves a new search set up based on entered values in the search fields with a dialog that allows you to define a name for the search setup.
Open configuration folder	Opens the directory in which the configuration files are stored in a file explorer window. Deleting these configuration files sets the dialogs back to default UI configurations.

The following search setups are defined by default;

- **Default Search**
- **Part ID** (Part tab)
- **Document ID** (CAD tab)

The search limit is set to **500** by default.

2.5.3 Results section

In this section, search results are shown in tabular form.

Sections in the Results section include:

- Metadata table with filter options
- Row items

Metadata table

Possible columns are described here Properties section/metadata table (p. 29).

Row items of metadata table

Model data is shown per row item in different columns. These selection methods are available:

- A selected check box in front of a row item will mark it for further processes in Innovator.
- A selected row item allows for context menu functions. You can either select one row item or multi-select several items by either **CTRL**+left-click, or a sequence with **SHIFT**+left-click.

2.5.4 Load options

The lower part of the dialog provides different load options, information about the number of found and selected objects and the load button.

You can select options that define how the model data is loaded to the local workspace.

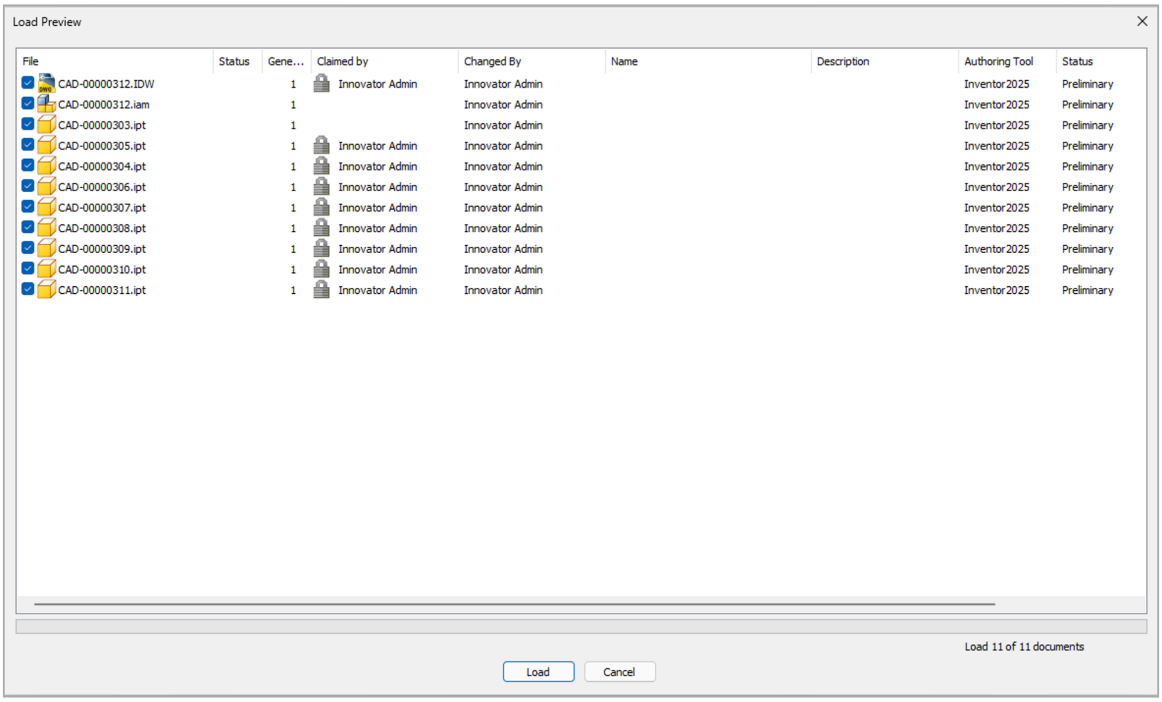
Option	Description
Use File Cache	If enabled, this option compares the local files with Innovator. If the files are the same, the integration doesn't download them again to save bandwidth; otherwise the local files are updated from Innovator. If disabled, the files are always downloaded from Innovator.
Rename files	If this option is selected, the integration renames all files during the check out to the local workspace.
Set reservation	If enabled, this option claims and reserves the model data automatically when loading. If disabled, the objects are not claimed and locked.
Update properties	This option triggers the **Update Properties** function after an object was checked out.
Load read-only	Checks out the files in read-only mode.
Load top-level only	This option is used to load the top element of a structure.
Show load preview	If this option is checked, the Load Preview is displayed after you have selected a model to load and clicked **Load <x> objects**
Resolution	■ **As-saved**: Checks out the structure as saved in Innovator. ■ **As-released**: Checks out released elements of the structure (else: current). ■ **Current**: Checks out the latest structure elements (head revisions).

After enabling or disabling required load options, press **Load [No.] objects** to load desired elements. To load multiple elements enable the check box in front of the desired elements and press **Load [No.] objects**. It is also possible to load only one document by **double-click** it.

2.6 Load preview

In this dialog, you can select which parts of the model structure you finally want to load. Checks ensure that nothing is overwritten locally without your explicit request.

This dialog appears if **Show Load Preview** is activated in the **Load options** section of the *Load dialog* and after you have selected a model to load and clicked **Load [No.] objects**.



Model data in the *Load Preview* is displayed in tabular form. The following information is provided:

Column	Description
File	Shows the file name of the Inventor model selected for load. Only selected files are downloaded. The checks performed in the background ensure that nothing will be overwritten by accident. You can always overrule the default action by enabling or disabling the check boxes in front of the file name. Selecting or deselecting files for download has no effect when loading a copy.
Status	Gives information about the status of the file regarding the Innovator information. There are four possible values: ■ - The file exists locally and has the same timestamp than the version within Innovator. File is not downloaded to preserve resources and enhance performance. ■ - The file exists locally and has been modified locally. File is not downloaded; local modified version is kept. ■ - The file exists locally but is older than the same generation of the file within Innovator. File is downloaded in order to update to the latest version. ■ Empty - File does not yet exist locally. File is downloaded.

Column	Description
Gen	Displays the generation of the file selected for load. In case it is not the latest generation, a warning is displayed in form of a yellow warning light in front of the generation version. The latest generation does not have a warning light. 
Claimed by	Shows the name of the user who has claimed the object for modification in Innovator.
Changed By	Shows the name of the user that made the last modification to the object in Innovator.
Name	Displays the name of the object in Innovator
Description	Displays the description of the object in Innovator
Authoring Tool	Shows the Inventor version in use.
Status	Shows Innovator workflow status.
Classification	Shows the Inventor file type.

Context menu

Right-click on a line item(s) in the *Load Preview* and you can select additional functions from the context menu.

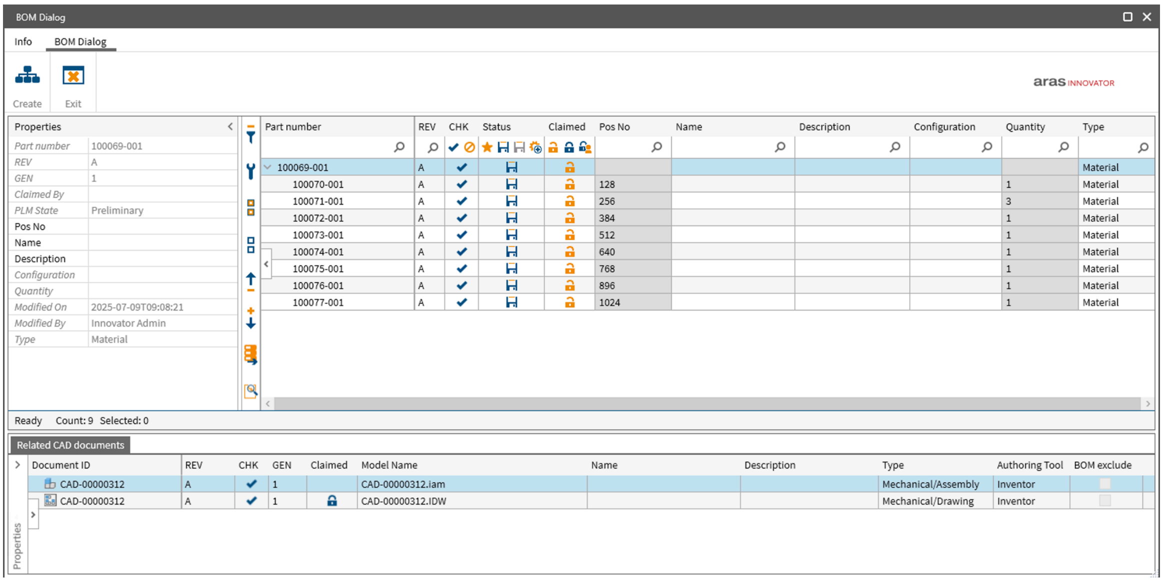
Function	Description
Claim Document	Reserves Innovator documents for currently selected CAD files
Unclaim Document	Removes reservation in Innovator of currently selected CAD files

2.7 BOM dialog

In this dialog, you can create parts and BOMs for your 3D models and edit part items.

The *BOM* dialog is displayed by using one of these functions within the integration menu:

- **Create Part and BOM**
- **Create Part and BOM (1st Level)**



The menu ribbon of the *BOM dialog* has two tabs

- Info tab (p. 28)
- BOM dialog tab (p. 25)

Available sections in the *BOM dialog*:

- Properties section (Properties section/metadata table (p. 29))
- Side bar (p. 28) (collapsible)
- Model and Related CAD documents sections (Model section (p. 25))

The *BOM dialog* also offers additional functionality in the Model section through;

- Context menu (p. 32)
- Filter options (p. 32)

The appearance of the *BOM* dialog can be changed by the system administrator; it does not necessarily have to appear like the one in the example above. See Installation and Administration Guide for more information.

2.7.1 BOM dialog tab

This section contains the main functions for creating parts and bills of material in Innovator.

Function	Description
Create	Starts the part and BOM creation process.
Exit	Closes the dialog without further actions.

2.7.2 Model section

This section shows the model data with its metadata in tabular form.

The model window contains two main panels:

- Parts panel
- Related CAD documents panel

Each of the panels has the following sections:

- **Properties section**
- **Side bar**
- **Metadata table**

Parts panel

In the Parts panel metadata table, columns with properties of the relation are highlighted in a different style, for example the Quantity column. Rows of new items are also highlighted in a different style. Items are set to read-only in case of

- no CAD document exists in Innovator to the local file
- the BOM exclude flag is activated or
- if it is an external position.

Read-only items are also highlighted in a different style, their context menu is disabled and properties cannot be edited.

External position
External positions are displayed in the BOM structure. They can be added via the Innovator client or a separate connector (such as ECAD). These elements are read-only

BOM exclude
The BOM exclude flag is a property of the related CAD document and is set in the *Save dialog*.

Related CAD documents panel

Displays related CAD documents of the selected part item in read-only and supports following context menu functions:

- Lock
- Unlock
- Get metadata
- Open in PLM

Metadata table

Possible columns are described here Properties section/metadata table (p. 29).

Row items of metadata table

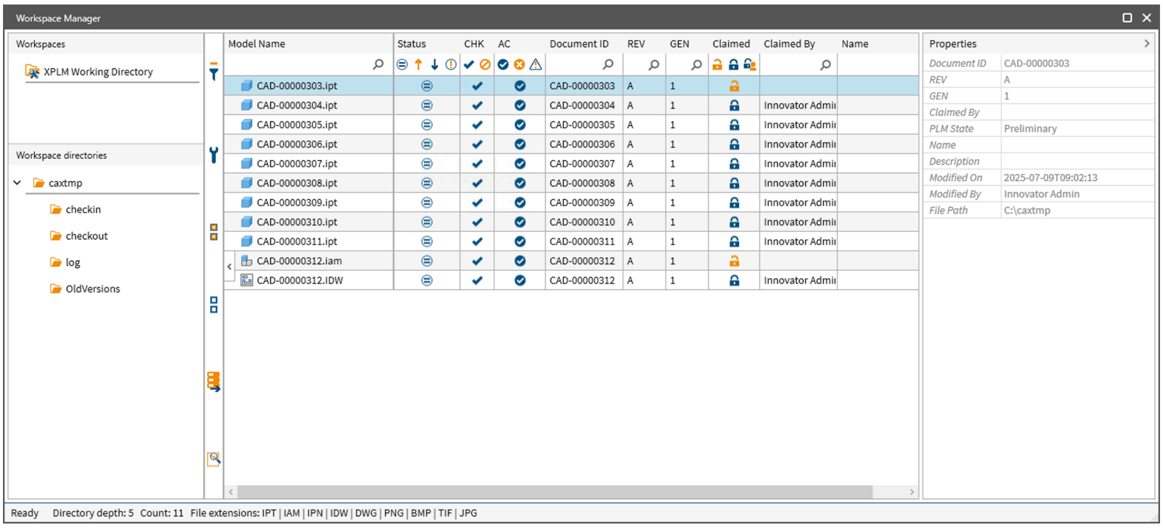
Model data is shown per row item in different columns. These selection methods are available:

- A selected check box in front of a row item will mark it for further processes in Innovator.
- A selected row item allows for context menu functions. You can either select one row item or multi-select several items by either **CTRL**+left-click, or a sequence with **SHIFT**+left-click.

2.8 Workspace manager

In this dialog you can quickly check the state of local files. The *Workspace Browser* dialog has primarily informational function and does not allow save or load.

The *Workspace Manager* dialog is displayed by **Innovator** > **Show Workspace** command within the integration menu.



Sections in the *Workspace Manager* dialog (from top and left to right):

- Workspace section (p. 27)
  - Workspaces
  - Workspace Directories
- Side bar (p. 28) (collapsible)
- Model and Property section (Properties section/metadata table (p. 29))
- Footer (p. 27) - Status line with number of available files in the model window plus the allowed file extensions

The *Workspace Manager* offers additional functionalities in the Model section through;

- Filter options (p. 32)
- Context menu (p. 32)

2.8.1 Workspace section

This section displays the existing workspaces.

The upper part shows the available workspaces (default and user-defined workspaces). By default, the standard workspace is `C:\caxtmp`, named *XPLM Working Directory*. The path can be changed using function **Select Workspace**.

The lower part shows the folder structure of the workspace which is selected in the upper part. When starting the workspace browser, the directory of the the active model tab is automatically expanded and its content is shown in the model window.

If you hover the mouse over the workspaces, the path is displayed.

For user-defined workspaces, See Installation and Administration Guide for more information.

2.8.2 Footer

The footer displays information about the number and file type of available files in the currently selected workspace.

The file types are divided in different groups. If one file type of a group is detected, all file types of the group are displayed. For user-defined workspaces, it is possible to define which file types should be displayed.

See Installation and Administration Guide for more information.

2.9 Common dialog sections

In this chapter, common sections of the XPLM dialogs are explained.

2.9.1 Info tab

In this tab you can enable functions for troubleshooting integration problems.

Also referred to as **XPLM Backstage area**. It contains following functions:

Option	Description
Version	Shows the version of the dialog.
Logging	Activates and deactivates the logging function of the dialogs and prompts for a location of the log file.
System messages	Activates and deactivates an additional area at the bottom of the dialog that shows system messages about results of user exits (information, warnings, aborts, failed checks).
Exit	Closes the dialog.

2.9.2 Side bar

This section can be shown or hidden and is located between the Properties and the Model sections. It contains functions to reset filter options, show/hide columns in the model section, select/deselect line items, and expand/collapse the model structure.

Function	Description
Clear filter	Clears all filter options in the column headers of the model window.
User profile	Shows a selection list for showing/hiding columns in the model window and to define the default view (structured or flat). To reset everything back to default, click **Reset profile**.
Select all documents	Selects all line items in the model window.
Deselect documents	Deselects all line items in the model window.
Collapse structure	Collapses the model structure (not applicable for flat view or in the *Workspace browser*).
Expand structure	Expands the model structure (not applicable for flat view or in the *Workspace browser*).
Data export	Exports the contents from the dialog to a file. This can be used to reproduce unexpected system behavior.

Function	Description
Show selected documents	Shows the data managed in the background of the dialog. The header of the document contains control data, the list of child nodes corresponds to the selected background data. You can search for (case sensitive) values in the document or export the complete document structure to the clipboard for further analysis.

2.9.3 Properties section/metadata table

This section shows all possible properties in the property grid and the metadata table. Following table shows all possible properties. Available properties can differ in the dialogs and depend on the one used.

Property	Description
AC	Shows accessibility of files in Innovator with regard to write privileges. Access conflicts might occur for documents that are released or protected in another way in Innovator. ■ - Document is accessible. File can be saved to Innovator. ■ - Document is not accessible. File cannot be saved to Innovator. ■ - Only write access. Files are not saved.
Authoring Tool	Displays the authoring tool of the file.
Authoring Tool Version	Shows the Inventor version.
BOM exclude	If check box is activated, no part and BOM is created for the related row element.

Property	Description
CHK	Indicates whether an object can be saved to Innovator, based on the user's privileges and the state of the object in Innovator. ■ - File can be saved to Innovator. ■ - File cannot be saved. Shows the failed checks and displays one of various icons indicating why the check failed in the tooltip. Possible errors include: □ - File is claimed by another user. □ - File is unclaimed by current user □ - There exists no object in Innovator for the collaboration data. User may be connected to the wrong database. □ - Innovator prevents updating the file. □ - File does not correspond to current generation. It indicates that a newer generation exists in Innovator. The latest generation does not have this warning dot. □ - Object is modified in Innovator. If one of these checks fails, affected objects are not preselected for further processes in dialog, even though they have been modified.
Claimed	Shows the claim status of the file. Possible values are; ■ - File is claimed in Innovator by current user. ■ - File is not claimed in Innovator. ■ - File is claimed in Innovator by another user.
Claimed by	Shows the name of the user who has claimed the object for modification in Innovator.
Configuration	Displays the name of the configuration.
Copy actions	Shows the selected copy actions. ■ - Copy Document ■ - Copy drawing ■ **Unselected** - Filter objects where no copy action is applied. ■ **Selected** - Filter objects where a copy action is applied.
Created By	Displays the name of user that created the file.
Created On	Displays the creation date.
Description	Displays the description of the object in Innovator
Document ID	Shows the Document object ID in Innovator.
File Path	Shows the local file path in the active working directory.
GEN	Shows the generation in Innovator of the currently loaded model.

Property	Description
Model Name	Shows the file name of the Inventor model with extension. For the model window section, the check box in front of the model name indicates whether the file is selected.
Modified By	Shows the name of the user that made the last modification to the object in Innovator.
Modified On	Shows the date and time the object was last modified in Innovator.
Name	Displays the name of the object in Innovator
New Document ID	Shows the document object ID of the newly created copy in Innovator. Available within the *Documents* and *Related drawings* panels.
New Part ID	Shows the part object ID of the newly created copy in Innovator. Only available within the *Related parts* panel.
Part ID	Shows the part object ID in Innovator.
PIC	Shows a thumbnail of the selected item. Move the mouse pointer over the image to enlarge it.
Pos No	Displays the BOM position numbers.
Quantity	Displays the quantity of the occurrences.
REV	Shows the object revision in Innovator.
State	Shows Innovator workflow status.
Status	Shows the status of the local file in Innovator. Possible values are: ▪ - File exists locally and has same time stamp as the version within Innovator. On saving, file is not uploaded to preserve resources and enhance performance. ▪ - File exists locally and is modified locally or is not known in Innovator. On saving, files are uploaded. ▪ - File exists locally but is older than the same generation of the file within Innovator. On saving, file is not uploaded; local modified version is kept. ▪ - File exists locally and is modified locally and in Innovator. File is not uploaded; local modified version is kept.
Status (BOM dialog)	Shows the status of the part creation in Innovator. Possible values are: ▪ - A new part is created in Innovator for the object. ▪ - Part exists in Innovator. ▪ - External position, no part is created or updated. ▪ - After selecting **Assign Part** from the context menu, it shows the status that a part is assigned.

Property	Description
Type	Shows the Inventor file type.
Unit	Available in the *Load dialog*, **Part** tab. Shows the units of the objects in Innovator.

2.9.4 Filter options

The dialogs offer additional functionalities to manage data. Columns of the model section can be filtered in the following way.

In case of a large file list, you can filter rows by columns.

In all columns with type `TEXT`, enter a search string under the column header then press **Enter** on your keyboard. Query results are displayed immediately.

You can also set filter condition by pressing yellow filter icon. This icon becomes visible after inserting a search string. Default condition is `Contains`.

Following conditions are available:

Condition	Description
Clear Filter	Removes all set filters and displays complete list of objects again.
Is equal to	Filters for exact match.
Is not equal to	Shows results which do not contain entered value.
Starts with	Name or value starts with the entered string.
Ends with	Name or value ends with entered string.
Contains	Searched name or value contains entered string.
Does not contain	Searched name or value does not contain entered string.
Is Empty	Filters for rows that do not contain any value.
Is not empty	Filters for rows that contain a value.

Columns with type `FilterImage` can be filtered by pressing one or more icons under column header. For example, Status, CHK, AC, and Claimed. Click on the image again to clear the filter

Columns with magnifier icon can be filtered by clicking on the magnifier icon. A selection dialog opens with available values. For example, ID, REV, Name and so on. When you select a value, the view is updated immediately. Press **Clear Filter** to remove all selections. Press **Exit** to close the selection dialog.

All filters are reset if you toggle between Structure and List view.

2.9.5 Context menu

Right-click on one or more selected items opens a context menu with additional functions.

Following table shows a list of available context menu functions. The available functions differ in the XPLM dialogs.

Function	Description
Select	▪ **Selected [CTRL+Space]** - Marks the file as selected. Check box in front of the **Model Name** is checked. ▪ **Recursive** - Marks the file and its underlying components as selected. Check box in front of the **Model Name** for all the selected files is checked.

Function	Description
Deselect	■ **Selected [CTRL+SHIFT+Space]**- Removes selection for the file. Check box in front of the **Model Name** is not checked. ■ **Recursive** - Removes selection for the file and all its underlying components. Check box in front of the **Model Name** is not checked for all deselected files.
Claim	■ **Selected [CTRL+L]** - Reserves Innovator documents for currently selected CAD files (only if an assignment exists). ■ **Recursive**- Reserves Innovator documents for currently selected CAD files and underlying components (only if an assignment exists).
Unclaim	■ **Selected [CTRL+SHIFT+L]** - Removes reservation in Innovator of currently selected CAD files (only if an assignment exists). ■ **Recursive**- Removes reservation in Innovator of currently selected CAD files and underlying components (only if an assignment exists).
Restore generation	Saves older generation as newest under specific conditions.
Get metadata	■ **Selected [CTRL+M]**- Gets information from Innovator and updates information in *Save Preview* for currently selected CAD files (only if an assignment exists). ■ **Recursive [CTRL+ALT+M]**- Gets information from Innovator and updates information in *Save Preview* for currently selected CAD files and underlying components (only if an assignment exists).
Open in PLM	**[CTRL+ALT+O]**- Opens Innovator Web Client and displays CAD Documents that are assigned to currently selected CAD file. (only if an assignment exists). This option does not work for multiple files.
Load Structure (1. Level)	Adds the first level children of the selected object to the Search dialog.
Load Structure (all)	Adds the all the children in the structure of the selected object to the Search dialog.
Copy dialog	Opens the *Copy* dialog.
New part number	Creates a new part in Innovator for the selected object(s) and displays it in the *BOM* dialog. It's only enabled for new part items.

Function	Description
Assign part	Assigns an existing part in Innovator for the selected object. After executing this function, the search dialog for parts opens to search for the desired part. Select the desired part and press **Load** to assign it. After the part is assigned, the status column changes accordingly. If a BOM structure exists for the assigned part, it is deleted. The BOM structure which is under the assigned part in the dialog is added. If a relation to an old CAD document exists it is deleted and the relation to the new assigned CAD document is created.

Innovator related functions are disabled if files are PLM-unknown.

2.10 Info dialog

This dialog provides key information about the integration's software components and active licenses.

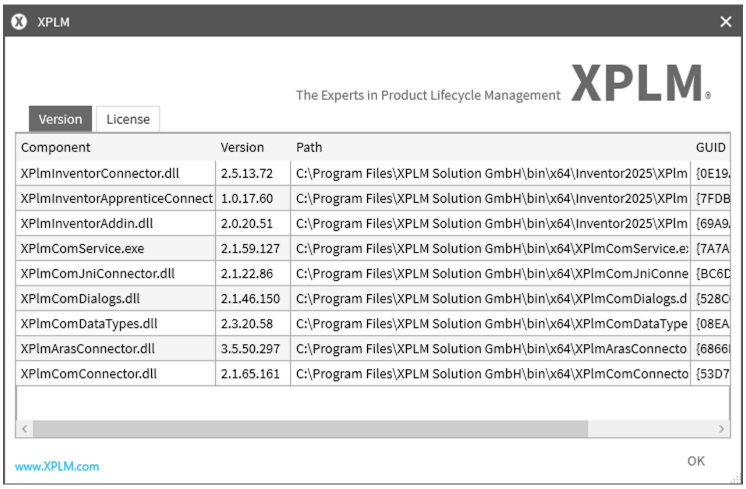
The *Info dialog* is displayed by the **Info** command within the integration menu.

The dialog has 4 tabs:

- Version tab
- License tab
- Properties tab - Displays all properties defined in `XPlmConnector.xml` files.
- Messages tab - Displays the resources defined in `XplmResource.xml` files.

Version tab

The **Version** tab displays a list of core components used by the integration.

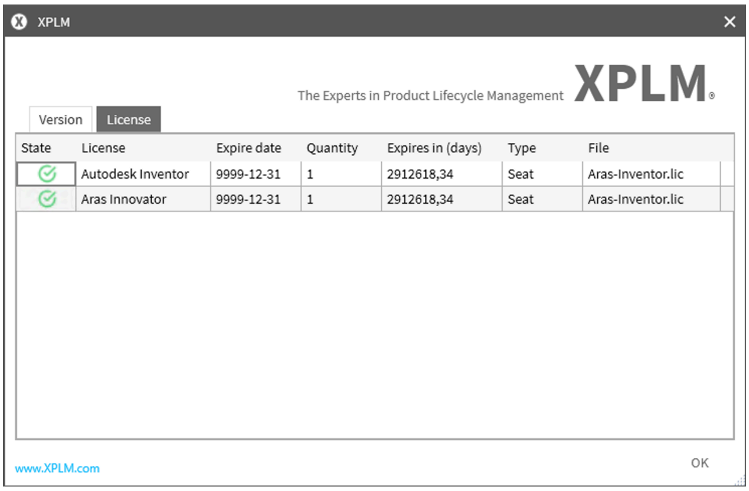


Each list entry provides the following details;

Column	Description
Component	Displays the name of the component.
Version	Displays the file version number currently in use.
Path	Displays the full path of the component on the local system.
GUID	Displays the GUID of the component.

License tab

The **License** tab shows information about the currently installed and active licenses used by the integration.



For each license entry, the following details are given;

Column	Description
State	Indicates whether the license is valid or expired. The status is indicated via icon color coded system; ▪ Green - licence is in good condition (not yet expired nor about to expire). ▪ Yellow - license is about to expire. ▪ Red - license has already expired.
License	Displays the name of the -license module.
Expire date	Displays the calender date on which the license expires.
Quantity	Shows the number of available license instances.
Expires in (days)	Shows the number of days remaining until the license expires.
Type	Specifies the license type.
File	Displays the name of the license file.

3 Usage

3.1 Starting the Integration

The Inventor connector for Innovator is operated directly from Inventor.

The integration functions are available and enabled by simply starting Inventor system after installation.

The **Innovator** menu is added in the ribbon bar within your Inventor system from which the integration functions can be executed.

3.2 Creating new Objects

The integration helps you to create new objects simultaneously in Inventor and Innovator.

About this task

The following integration menu functions create new objects, saves them locally and to Innovator with new CAD Documents ID and opens the objects in Inventor for modification:

- New Part
- New Assembly
- New Drawing

The created objects are automatically claimed after creation for the current user.

This use case describes how to create a new part file using the **New Part** function.

Procedure

1. Launch Inventor and click **Innovator** > **Connect**

   ➔ Connection to Innovator established.

2. Click **Innovator** > **New Part**

   ➔ The *Save dialog* is opened, the part is preselected for save.

3. **Optional:** In the *Save dialog*, enter values in the fields **Name** and **Description** within the Model window or the Properties window.

4. Click **Save** within the *Save dialog*.

Result

A new part is created, saved to Innovator with new CAD Document ID and opened in Inventor. The file is also saved to the local workspace with its corresponding .CLB file.

What to do next

Modify the part in Inventor as desired.

3.3 Save to Innovator

3.3.1 Saving to Innovator via Save Dialog

The integration helps you save model data consistently from Inventor to Innovator.

Before you start

Launch Inventor and create or open a model that is NOT known in Innovator.

About this task

This use case describes how to save new and modified Inventor files to Innovator via the **Save Dialog**.

Procedure

1. Press **Innovator** > **Save**

   ➔ Save dialog opens. All files are selected for saving. Column *Model Name* is automatically filled with object name from Inventor. Columns with Innovator related information are not filled.

2. **Optional:** Add **Description** and **Name** by double-clicking appropriated cells in the Model window or using the Properties window. These values are mapped to corresponding fields in Innovator during save process.

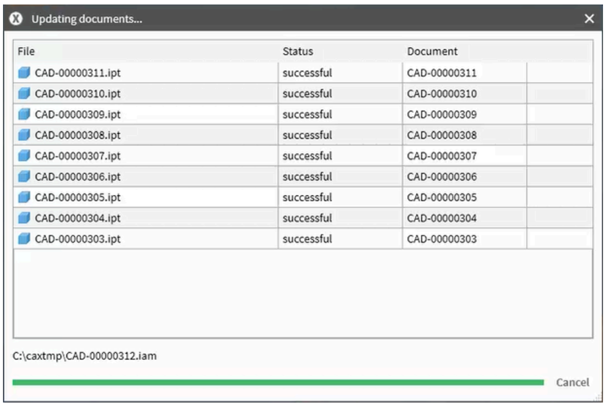
3. **Optional:** If the Inventor model has a structure, activate the **Update BOM** button to create BOM in Innovator after save.

4. **Optional:** Activate the **New generation** button if new version should be created in Innovator after save.

5. Press **Save** button within the Save dialog.

   When saving for the first time, Document Numbers are assigned automatically to each model.See Document number assignment (p. 38) for more information.

   Save progress dialogs are displayed and automatically closed when the save process is finished.

   ➔ Save process starts and finishes without errors. Files and information are uploaded to Innovator.

6. Return to Inventor and modify an opened object for example the top element and a child element.

7. Press **Innovator** > **Save**

   ➔ Save dialog opens. Modified files are preselected for saving. Innovator related information for example Document Number, REV, GEN, PLM State and so on are displayed for all files. Furthermore, the *Save* dialog provides information about Claim status and corresponding user.

8. **Optional:** To update PLM related information for all files, perform a **right-click** on top element and select **Get Metadata** > **Recursive** from context menu.
   If user needs information about a specific file, perform a **right-click** on desired file and select **Get Metadata** > **Selected**. Multiple selection is also possible.

9. **Optional:** Select all files for saving. Perform a **right-click** on top element and select **Select** > **Recursive** from context menu. Or select **All** from the *Selection* fly-out menu (upper left area of the dialog).

   ➔ All files are selected for saving. Non-modified files are marked with an equal sign in the **Status** column. Means that nothing is uploaded to Innovator during the save process.

10. Press **Save** button.

Result

Save process starts and finishes without errors. Files and information are uploaded to Innovator.

Related links
Save dialog (p. 15)

3.3.2 Model Status handling

Using the **Show selected documents** function available in the Side Bar of the *Save dialog*, displays the `XPLMDocument` which shows the data managed in the background of the dialog.

Model status is part of this data managed in the background of the *Save dialog*.

When reviewing model status in the *Save dialog*, two fields are relevant:

1. **ModelModified**
   - Indicates whether the model has changed based on **timestamp** only.
   - It is used to determine if the model requires saving due to actual file changes.
   - If this field is `true`, the model has changed since the last save and should be saved.
2. **ModelModidfiedCAD**
   - Indicates whether the CAD system reports the model as internally modified via the API call `UF_PART_is_modified`.
   - It is useful for diagnosing CAD-specific behaviors where internal flags differ from timestamp changes.

Models will only be marked for save if timestamps have changed. Inventor internal flags are now tracked separately in `ModelModifiedCAD`.

Example Scenario

If the Inventor save button is pressed locally but the modified flag is not reset by Inventor, `ModelModifiedCAD` may be set to `true` while `ModelModified` remains `false`. In this case, the model will not be saved because no timestamp change occurred.

3.3.3 Document number assignment

When 2D models are saved to Innovator, they are automatically assigned the same document number as their associated 3D models by default.

Introduced or updated with product version 4.5.0.

If a 3D model is already PLM-known, any newly created drawing related to that model will get the same Document Number as the 3D model.

Handling multiple drawings for the same 3D model

If multiple drawings are created for the same 3D model, a numerical suffix (for example `_01, _02, _03 and so on`) will be appended to the drawings' Document Number in Innovator to ensure uniqueness.

Example:

- 3D Model Document Number : `CAD-00000126`
- First Drawing Document Number : `CAD-00000126`
- Second Drawing Document Number : `CAD-00000126_01`
- Third Drawing Document Number : `CAD-00000126_02`

Handling a drawing with multiple 3D models

If a drawing has multiple related 3D models, the drawing is assigned the same Document Number as the first model.

Example:

- First 3D Model Document Number : `CAD-00000126`
- Second 3D Model Document Number : `CAD-00000127`
- First Drawing Document Number : `CAD-00000126`
- Second Drawing Document Number : `CAD-00000126_01`
- Third Drawing Document Number : `CAD-00000126_02`

Rules for Document Number assignment

1. The first drawing of a 3D model gets assigned the same Document Number as the model .
2. Subsequent drawings of the same model get incremental numerical suffixes starting with `_01,` `_02, _03 and so on.`.
3. The Document Number of the 3D model remains unchanged regardless of how many drawings are associated with it.
4. If a drawing has more than one related 3D model, then the drawing is assigned the same Document number as the first 3D Model.

3.3.4 Automatic PDF generation

The Inventor connector has been enhanced to automatically generate PDF files when drawings are saved to Innovator.

Introduced or updated with product version 4.5.0.

When a drawing is saved to Innovator, a PDF file is automatically created

- Only one PDF file is created. If a drawing has multiple sheets, one PDF page is created for each drawing sheet.
- The PDF name matches the CAD Document number of the drawing with format `{DrawingDocumentNumber}.pdf`.
- The PDF file is attached to the **Viewable file** property on the drawings CAD Document.
- The PDF file is automatically updated when drawing is revised and saved to Innovator.

3.3.5 Customizing the save dialog

You can customize the look of the *Save dialog* to display the columns you want in the order you wish.

About this task

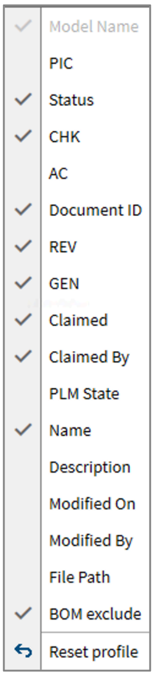
This use case describes how to customize the *Save dialog*.

Procedure

1. Launch Inventor and open or create a new file.
2. Press **Innovator** > **Save**

   ➔ *Save dialog* opens.

3. On the *Save dialog*, click the right facing arrow between the model window and properties window to expand the side bar.
4. In the side bar, click on the **User profile** icon.

   ➔ A list of available columns appears.
5. **Optional:** Click on the column names in the displayed list to either select or deselect them.

   ➔ Selected columns (with a blue tick icon) are added and deselected columns are removed immediately from the *Save* dialog and saved to the profile.
6. Click on the **User profile** icon again to hide the list.
7. **Optional:** Change the order of the displayed columns in the *Save dialog* by dragging and dropping them to desired positions.
8. **Optional:** Change size of columns in *Save dialog* by positioning the mouse pointer over a resizeable border of a column header, **left click** and move the mouse. Release mouse if desired size is reached..

   ➔ Columns have desired size. The information is saved in currently active user profile.

   Columns with icons are not resizeable.
9. **Optional:** Adjust the size of the *Save dialog* by positioning the mouse pointer over a resizeable border, **left click** and move the mouse. Release mouse when desired size is reached.

   ➔ *Save dialog* has desired size. The information is saved in currently active user profile.

Result

Save dialog is adjusted as desired.

What to do next

If the *Save dialog* should appear in its default version, select **User Profiles** > **Reset profile** from the side bar.

Function **Reset profile** should only be used, if system administrator has changed configuration.

3.3.6 Restoring older generation as newest

Use the **Restore generation** context menu command in the *Save dialog* to create a new generation from an older generation of a CAD Document.

Before you start

Verify that you have a valid CAD Document of an assembly in Innovator not claimed by another user In multiple generations

Inventor is running and you are connected to Innovator with a user with elevated privileges.

Procedure

1. Load the CAD Document to Inventor in an older Generation.
2. Claim the structure of the assembly (**Claim(Structure)**).
3. Press **Innovator** > **Save**
4. In the *Save Dialog*, select one or more objects that are not in the newest generation.
5. Right-click and choose **Restore generation** from the context menu
   ➔ Confirmation message `Do you want to restore selected older generation(s) as newest?` is displayed.
6. **Option:** Click **No** to Cancel.
7. **Option:** Click **Yes** to proceed.
   ➔ The systems performs validation checks;
   - If any selected object is already in the latest generation, a message is displayed `Following objects cannot be restored: CAD-XXXXX (Cannot restore: is current generation)`
   - If all checks pass, a new generation is created for all selected items. *Save Dialog* updates automatically (Status, Generation) and selected rows are marked for save.
8. Press **Save** button within the *Save dialog*.
   ➔ Save process starts and finishes without errors. Files and information are uploaded to Innovator.
   If you click **Exit**, the new generation remains in Innovator with CAD data from the previously newest generation.

Result

A new generation is created for each selected item.

Local CLB files are updated to point to the new generation.

Related links
Context menu (p. 32)

3.4 Copy CAD documents

This section provides detailed information on how to use the *Copy dialog*.

The *Copy dialog* allows you to copy Innovator CAD Documents such as 3D models (for example, Mechanical\Parts and Mechanical\Assembly) while configuring key properties of the copied items. The *Copy dialog* also allows you to copy associated drawings and parts of the selected documents. You can define new property values, use automatic item number generation and maintain structural consistency across the copied data.

Structure and display

The *Copy dialog* always displays the complete Innovator structure of the selected root element.

You can configure how the structure is queried and displayed through a configuration file. See Installation and Administration Guide for more information.

Related Drawings and Parts stored in Innovator are shown in separate dedicated grids on the *Copy dialog*. These related items appear only if their parent document is selected for copying. You can also choose to include them in the copy and define their properties.

Copy selection and property configuration

Select documents and related items for copying by checking the check box in the grid or using the context menu. Multi selection of items is supported.

- When a child document is selected, all parent documents in the hierarchy above it are also automatically selected for copy.
- You cannot deselect a parent document if any of its children are selected for copying.
- If a document is deselected from copying, all related drawings and/parts that were selected for copying will also be automatically deselected.

You can configure the grid columns to enable editing of properties for the new copies. See Installation and Administration Guide for more information.

A default document/part number is automatically generated for each new document.

Committing the copy

Once you have finalized all selections and property configurations, click the **Commit** button to create the copied documents in Innovator and apply the defined properties.

3.4.1 Copy during load

You can copy documents and their associated drawings or parts during the load process.

Before you start

Verify that you have a valid CAD document in Innovator not claimed by another user.

Inventor is running and you are connected to Innovator.

About this task

This use case describes how to copy documents during the load process.

Procedure

1. In Inventor, press **Innovator** > **Load from Document**.

   ➔ The *Load dialog* opens and the **CAD** tab is displayed.

2. Query for the document object you want to copy by entering the document ID in the **Document Number** field and clicking on **Search**.

   ➔ The queried document is shown in the results window.

3. Right-click on the file in the results section and select **Copy dialog** from the context menu.

   ➔ The *Copy dialog* opens.

   You can only initiate the *Copy dialog* from 3D models! If the queried document is a drawing, the following message is displayed to warn you of the conflict:

   > The selected document has the Aras classification 'Drawing'. Documents of this classification are not allowed. Please start the copy action from a 3D model.

4. On the *Documents panel* of the *Copy dialog*, select the documents to copy by manually marking the check boxes in front of the **Document Number** or via context menu **Select** command.

   ➔ The **New document number** field is automatically filled with the new generated document number for selected documents.

5. **Optional:** Select the related drawings and/or parts to copy in the corresponding grids.

   ➔ The **New document/article number** field is automatically filled with the new generated number respectively.

6. Press **Commit** in the *Copy dialog* .

   ➔ The *Copy dialog* and *Load dialog* are closed as the copy process starts and finishes.

Result

The native files of the copied CAD documents are downloaded from Innovator and opened in Inventor.

Related links

Copy dialog (p. 17)

3.4.2 Copy from integration menu

You can launch the *Copy dialog* directly from the integration menu to copy documents and their associated drawings or parts.

Before you start

Launch Inventor and open a model that is known in Innovator.

Ensure you are connected to Innovator.

About this task

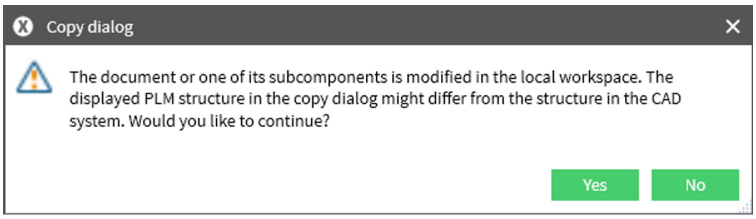
This use case describes how to copy Innovator documents directly from Inventor.

Procedure

1. In Inventor, press **Copy** from the integration menu.

   ➔ The *Copy dialog* opens displaying the full structure of CAD Document .

   - You can only initiate the *Copy dialog* from 3D models!
   - If the model is locally modified, the following warning is displayed;

     - Click **Yes** to continue copy without local modifications.
     - Click **No** to cancel the process.

2. On the *Documents panel* of the *Copy dialog*, select the documents to copy by manually marking the check boxes in front of the **Document Number** or via context menu **Select** command.

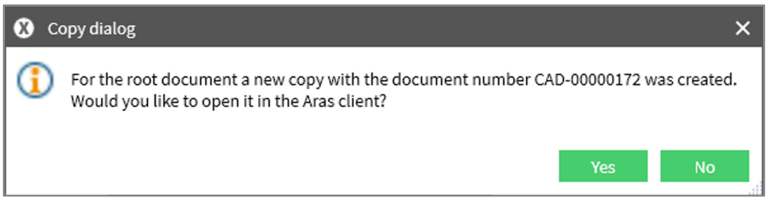
   ➔ The **New document number** field is automatically filled with the new generated document number for selected documents.

3. **Optional:** Select the related drawings and/or parts to copy in the corresponding grids.

   ➔ The **New document/article number** field is automatically filled with the new generated number respectively.

4. Press **Commit** in the *Copy dialog* .

   ➔ The *Copy dialog* closes and a pop up asking if you would like to open the new copy in Innovator is displayed.



   - Click **Yes** to open the new CAD Document in Innovator web client.
   - Click **No** to finish the process without opening the new CAD Document in Innovator.

Result

A duplicate of the selected document for copy is created in Innovator with new CAD Document number and configured properties.

3.4.3 Related Item handling and autoselect behaviour

Autoselect behaviour

Autoselect is triggered only when selecting CAD Documents, and it works recursively up to the root item. It is not triggered when seleccting parts or drawings.

Auto select is enabled out-of-the-box (OOTB) in the file in the transactions `DrawingConfig` and `PartConfig`, Field name `AutoSelect`.

Standardized Item Numbering

New index-based standard item numbers and Document numbers are introduced for parts and drawings associated with a 3D CAD Document.

Example numbering scheme for CAD-00001:

- CAD-00001
- CAD-00001_01
- CAD-00001_02

This numbering is zero-based and follows the behaviour of the Innovator method `GetUniqueItemNumber`.

Metadata binding for copied items

When copying items in the Documents, Drawings and Parts tree list views, the following Innovator fields are bound to `UpdatePropertiesCopy`;

- **Name**
- **Description**

The default values are derived from the original items' structure metadata. These fields are **editable** and their values are carried over OOTB during the copy operation.

Standard part documents

For CAD documents marked as **Standard** in Innovator,

- All related parts and drawings are excluded from copy (greyed out in the *Copy dialog*) regardless of whether they are marked as **Standard** or not.

Viewable file handling

During the copy process, any Viewable files for example PDF files associated with the original CAD Documents will be removed and will not be displayed under the copied Documents.

3.5 Load to Inventor

3.5.1 Load from document

The integration helps you load CAD Documents from Innovator database to the local workspace and optionally open them in Inventor.

Before you start

Verify that you have a valid CAD Document in Innovator not claimed by another user.

Inventor is running and you are connected to Innovator.

About this task

This use case describes how to load a CAD Document to Inventor via the **Load from Document** function.

Procedure

1. In Inventor, press **Innovator** > **Load from Document**

   ➔ The *Load dialog* opens and the **CAD** tab is displayed. The Results window is not active. The Search Criteria **Authoring Tool** already contains a value corresponding to the authoring tool in use.

2. **Optional:** Define Search criteria by entering values in the available fields.

   ➔ Search criteria defined.

3. Click **Search** within the *Load dialog*.

   ➔ The Results window is activated and is filled with the query results. If no search criteria are defined, this will by default return all entries from the database until the defined search limit is reached.

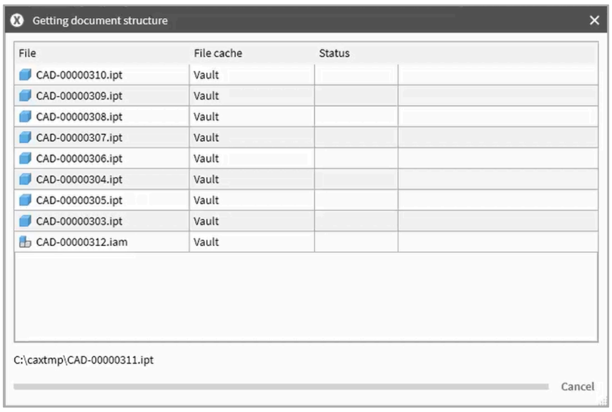
4. **Optional:** Activate the **Show load-preview** check box in the **Load options** section and set other Load options as desired.

   ➔ Load options set

5. Select the Documents to load by manually marking the check boxes in front of the Model Names or context menu.

6. **Optional:** Claim the Documents using the context menu function **Claim**.

7. Click **Load [No.] objects**.
   A Load progress dialog is displayed and automatically closes.



➔ The *Load Preview* opens. The list view is filled with corresponding Innovator and Inventor related information.

8. Select the files to load by manually marking the check boxes in front of the File Names.
9. Click **Load**.

Result

The native files of the CAD Documents are downloaded from Innovator and opened in Inventor depending on the load options selected.

Related links
Load dialog (p. 19)
Load preview (p. 23)

3.5.2 Load from client

You can also initiate the load process directly from the Innovator Web client.

Before you start

Verify that you have a valid CAD Document in Innovator not claimed by another user.

Inventor is running and you are connected to Innovator.

About this task

This use case describes how to load a CAD Document to Inventor via the **Load from Client** function.

Procedure

1. Open the Innovator Web Client and navigate to the desired CAD Document.
2. Click **More** > **Load to CAD**

   ➔ A pop is displayed informing you that the CAD item has been successfully added to the Load queue.
3. Return to Inventor and click **Innovator** > **Load from Client**

   ➔ The *Load dialog* opens and the **Load from client** tab is displayed. The Results window lists all the documents that have been added to the load queue.
4. **Optional:**Set the desired **Load options** by manually marking the check boxes in front of each option.

   ➔ Load options set.
5. Select the Documents to load by manually marking the check boxes in front of the Model Names or using context menu functions.

6. Click **Load [No.] objects**.
   A Load progress dialog is displayed and automatically closes when the load process is finished.

Result

The native files of the CAD Documents are downloaded from Innovator and opened in Inventor depending on the load options selected.

Related links
Load dialog (p. 19)

3.5.3 Load from part

The integration helps you load CAD structures from Innovator starting from a Part assignment.

Before you start

Verify that you have a valid CAD Document with Part assignment in Innovator not claimed by another user.

Inventor is running and you are connected to Innovator.

About this task

This use case describes how to load a CAD structure to Inventor via the **Load from Part** function

Procedure

1. In Inventor, press **Innovator** > **Load from Part**
   ➔ The *Load dialog* opens and the **Part** tab is displayed.
2. Enter the Part ID in the **Part Number** field and click **Search** within the *Load dialog*.
   ➔ The Results window is activated and is filled with the query results.
3. **Optional:** Set other Load options as desired.
   ➔ Load options set
4. Select the Part object to load by manually marking the check boxes in front of the Part Number or context menu Select function.
5. **Optional:** Claim the Part using the context menu function **Claim**.
6. Click **Load [No.] objects**.
   ➔ A Load progress dialog is displayed and automatically closes when the load process is finished.

Result

The native files of the assigned Part object are downloaded from Innovator and opened in Inventor depending on the load options selected.

When using **Load from Part** with **Resolution** set to **Current**, the top-level CAD document may not be loaded as current if it is fixed. This behavior is by design and results from the structure resolution logic, which does not affect fixed top elements.

3.5.4 One-Click load from Innovator

The connector has been upgraded with **One-Click Open** feature that allows you to load a CAD Document from Innovator directly into Inventor with a single click.

Before you start

Verify that you have a valid CAD Document in Innovator not claimed by another user.

Inventor is running and you are connected to Innovator.

About this task

Introduced or updated with product version 4.7.0.

The **One-Click Open** button is available in the Innovator browser interface when viewing a CAD Document.

Procedure

1. Open the Innovator Web Client.
2. Navigate to the desired CAD Document.
3. Click **One-Click** > **Open**

   ➔ A Load progress dialog is displayed and automatically closes when the load process is finished..

Result

The native files of the CAD Documents and corresponding CLB files are downloaded from Innovator and placed into the local workspace directory. The files are then automatically opened in Inventor.

3.6 Access Control Management

3.6.1 Claim and unclaim files

You can claim or unclaim a single object or the entire object structure in Innovator.

Claim and **Unclaim** functions are available in several locations of the integration:

- the integration main menu
- the *Save dialog*
- the *Search dialog*
- the *Load Preview*
- the *Workspace dialog*
- the *Copy dialog*

Verifying the claim status

You can verify the Claim status of an object by checking the **Claimed** and **Claimed By** columns of the integration dialogs.

The **Claimed** column shows the claim status of files with the following possible values:

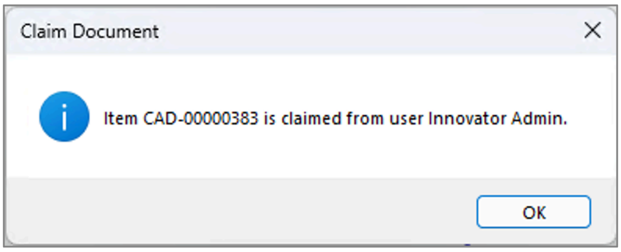
- - File is claimed in Innovator by current user.
- - File is not claimed in Innovator.
- - File is claimed in Innovator by another user. You cannot Unclaim files that have been claimed by other users.

If files are claimed, the **Claimed By** column displays the user that has claimed the files.

You can also confirm the Claim status of files in Innovator Web Client by checking the **Claim/Unclaim** flag. Claimed objects have a green flag.

Unsuccessful claim

In the event that you cannot claim an object in Innovator (for example, due to another user having already claimed the object in Innovator), an error message will be displayed warning you of the conflict.



3.6.2 Reserving files in Innovator via save dialog

Before you start

Launch Inventor and create or open a model that is NOT known in Innovator.

About this task

This use case describes how files can be claimed and unclaimed in Innovator via the *Save* dialog.

Procedure

1. Press **PLM** > **Save**.

   ➔ *Save* dialog opens. All files are selected for saving. Column *Model Name* is automatically filled with object name from Inventor. Columns with Innovator related information are not filled.

2. Press **Save** button within the *Save* dialog.

   When saving for the first time, Document Numbers are assigned automatically to each model.

   ➔ A Save progress dialog is displayed and automatically closes when the save process is finished. Save process finishes without errors. Files and information are uploaded to Innovator.

3. Reopen the *Save* dialog by pressing **Save** in the integration menu.

   ➔ *Save* dialog opens. Innovator related information for example Document Number, REV, PLM State and so on are displayed for all files. Furthermore, the *Save* dialog provides information about Claim status and corresponding user in the **Claimed** and **Claimed By** columns.

4. **Optional:** To Unclaim all files in the structure, perform a **right-click** on top element and select **Unclaim** > **Recursive** from opened context menu.
   If you want to Unclaim a specific file, highlight it in the model window, perform a **right-click** and select **Unclaim** > **Selected**. Multiple selection is also possible.

   ➔ All files in the structure are unclaimed. Pressing **Save** is not necessary for this action.

5. **Optional:** To Claim all files in the structure, perform a **right-click** on top element and select **Claim** > **Recursive** from opened context menu.
   If you want to Claim a specific file, highlight it in the model window, perform a **right-click** and select **Claim** > **Selected**. Multiple selection is also possible.

   ➔ All files in the structure are claimed. Pressing **Save** is not necessary for this action.

3.7 Property exchange

Properties (also known as parameters, attributes, or metadata) are information stored as text strings that are associated with Inventor and Innovator data. The connector supports bi-directional transfer of properties between Inventor and Innovator.

3.7.1 Property mapping

The connector supports mapping of properties to and from Innovator and Inventor.

When mapping from Inventor to Innovator, there are two types of properties that can be mapped:

- **System properties** - are not directly defined by the user such as the file name and the Inventor software version number which can be saved as properties in Innovator.
- **User properties** - are defined by the user using the Inventor custom file properties

System properties and user properties are mapped into Innovator Documents and Part objects as part of the Save process.

When working with drawings, the **Update Title Block** command is used to map Innovator properties into Inventor. This updates properties just for the current drawing, not all subordinate models. To use properties within a Title Block, you need to define the text notes to be linked to properties, either within the drawing or within the part or assembly referenced on the drawing.

See Installation and Administration Guide for more information.

3.7.2 Updating properties

The integration helps you map properties from Innovator to Inventor.

Before you start

Prepare a model file with three level structure in Inventor.

Inventor is running and you are connected to Innovator.

About this task

This use case describes how to map name attributes from Innovator to Inventor using the **Update Properties** functions.

Procedure

1. Open the prepared model in Inventor.

   ➔ Inventor displays the test model.
2. Save the test model and its complete structure to Innovator via the Save dialog.

   ➔ Model file is successfully saved to Innovator.
3. Click **Innovator** > **Display Document** from the integration menu.

   ➔ The Innovator Web client is opened displaying the CAD Document of the top model.
4. In the Innovator Web client, change the **Name** attribute value of the top model's CAD Document to a unique value. Then do the same for any child elements in the structure.

   ➔ **Name** attribute in Innovator has been changed for all CAD Documents in the structure.
5. Return to Inventor and click **Current**

   ➔ A progress bar may be displayed for a very short time, otherwise nothing discernible happens.
6. Open the Inventor file properties of the root model.

   ➔ A *Name* property has been added to the Custom Properties with the unique value added in step 4. In addition, the following properties from Innovator are also added

   - Document Number
   - Document ID
   - Document Type
7. Return to Inventor and click **Structure**

   ➔ A progress bar may be displayed for a very short time, otherwise nothing discernible happens.
8. Open the Inventor file properties of the child elements of the test model.

   ➔ A *Name* property has been added to the Custom Properties with the unique value added in step 4. In addition, the following properties from Innovator are also added:

- Document Number
- Document ID
- Document Type

Result

Properties have been successfully mapped from Innovator to Inventor. See Installation and Administration Guide for more information.

Related links
Saving to Innovator via Save Dialog (p. 37)

3.8 Material and BOM management

3.8.1 Assigning parts

The connector provides several ways for assigning Part objects to a CAD Documents in Innovator.

Automatic part assignment

You can assign Parts automatically during the Save process by activating the **Update BOM** button on the Save dialog.

Since Release 4.5.0, if **UpdateBOM** is enabled while saving drawings with structure to Innovator, the drawing document objects are attached to the same Innovator Part as their underlying 3D model Document objects.

Manual part assignment

You can assign Parts manually using the **Create Part** and **Create BOM** functions on the integration menu.

Deleting a part assignment

Part assignments relationships can only be removed from Innovator Web Client.

To remove a Part Assignment:

1. open the Part in the Innovator Web Client
2. go to the tab **CAD Documents**
3. click **Edit**
4. highlight the CAD Document and click **Delete Part CAD**
5. Click **Save** > **Done**

The original CAD Document may have to be closed and reopened.

3.8.2 BOM creation

BOM creation is used to create or update Innovator BOM tables based on Inventor structures.

The connector's BOM publishing process has three main steps;

1. **Save** - This step saves the Inventor structure into Innovator, using the **Innovator** > **Save** command.
2. **Create Parts** - This step links each CAD Document object to a corresponding Part object. The Part relationship is displayed on the CAD Document in the Innovator Web Client.
3. **Create BOM** - This step creates or updates the BOM structure for all the assigned Parts, based on the corresponding Inventor structure.

If necessary, manual BOM edits can be made in Innovator Web Client. Manual BOM edits are tracked independently and do not change upon subsequent BOM publish updates.

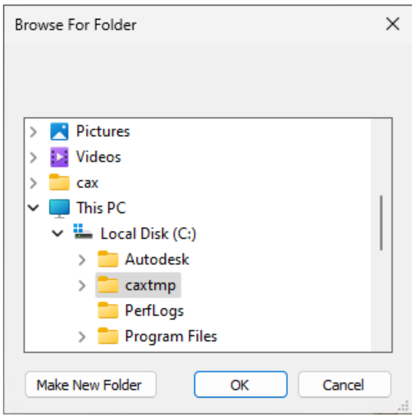
3.9 Workspace Management

The integration uses a local working directory to store Inventor files. The connector synchronizes these local files with Innovator. You can have multiple workspaces in parallel in order to separate the local data, for example when working on different projects at the same time.

3.9.1 Select workspace

The Inventor connector enables you to choose any local directory as the current local working directory with the function **Select Workspace**.

This function launches a directory selector window from which you can browse the available directories and select the desired folder and click **OK**. The new workspace is set immediately.



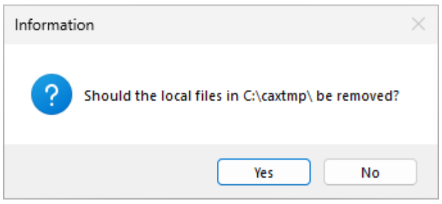
Subsequent load operations download data from the Innovator database into the selected workspace directory.

It is strongly recommended to use this function only if nothing is loaded in Inventor. Changing the workspace directory while having an Innovator-known file loaded in Inventor, may result in the file getting saved to Innovator again to with a new CAD Document number. This might destroy existing structures.

3.9.2 Remove local files

The Inventor connector provides an easy way of cleaning the local temporally workspace with the function **Remove Local Files**.

After selecting this function, a confirmation message appears. Local files are removed upon confirmation. The confirmation message also appears when closing the Inventor system.



The files to be deleted can be configured. See Installation and Administration Guide for more information.

If files are deleted, the corresponding .CLB files are also removed.

3.9.3 Collaboration files

Collaboration files (.CLB files) are files that contain meta information of Inventor files managed by Innovator.

Collaboration files are automatically created by the integration and used to link a Inventor file to its corresponding Innovator object. A .CLB file always has the same file name as its source file, along with the file extension **.CLB**.

Whenever a file is saved or downloaded from Innovator, using the integration, a new collaboration file is created and saved in the same directory as the transferred file. If a collaboration file is deleted, the link between the file on the local workstation and the corresponding object in Innovator is broken. Due to this breakage, the integration cannot update the Innovator object any longer and treats the source file as unknown to Innovator, thus creates a new Innovator object when the source file is saved again.

3.10 Inventor specific functionality

3.10.1 Thumbnail support

The connector extracts thumbnails from the Inventor native file.

The Inventor system stores thumbnail views into the Inventor binary data that are extracted using the same routines used by Windows Explorer.

The connector does not generate the thumbnail content but retrieves it from Inventor files. Please refer to the documentation of your Inventor system on how to enable thumbnail generation in the Inventor files.

The thumbnails are only held locally and are shown inside the Save, Load and Workspace dialogs in small icons. A bigger fly-out is shown on the small icon by hovering your mouse over the image. In the Load Preview the thumbnail from Innovator is downloaded if no local thumbnail is available. In all other dialogs, only thumbnails generated locally are used.

See Installation and Administration Guide for more information.

3.10.2 Handling assembly configurations

The Inventor integration supports the creation and update of bill of materials (BOM) of assembly configurations.

In Innovator, the assembly is saved to one CAD Document but for each configuration of the assembly, a corresponding Innovator Part can be created. Since different configurations may have different bill of materials, each configuration Part maintains its own material list.

To create such a configuration part, save the current Inventor assembly configuration to Innovator then execute function to create the parts or create the BOM.

You must create parts and BOM for each assembly configuration separately to create the corresponding part for the configuration.

The results can be inspected in Innovator on the CAD Document of the assembly where the configurations are shown in the document-part-relation. For every configuration one Part is created.

The Parts are stored under the **Parts** tab of the assembly CAD Document. By double-clicking the single Parts, the corresponding bill of materials is shown.

Using only the **Create Parts** function from the integration menu only assigns Parts to the corresponding configuration Part but a BOM is not created in Innovator. You must use the function **Create Part + BOM** for each configuration-part to have its own BOM.

3.10.3 Handling recursive structures

The connector is is extended to handle recursive structures when saving.

Since Autodesk Inventor 2022, assemblies with recursive structures can only be saved to Innovator if `Master` is activated for the assembly.

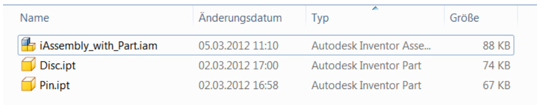
Refer to About Model State for more information.

3.10.4 Handling iAssemblies

3.10.4.1 iAssembly with part (assembly variance)

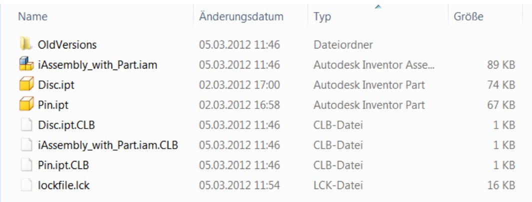
Example:

Figure 3: Local file cache: Before saving to Innovator (1 iAssembly, 2 parts)



Name	Änderungsdatum	Typ	Größe
iAssembly_with_Part.iam	05.03.2012 11:10	Autodesk Inventor Asse...	88 KB
Disc.ipt	02.03.2012 17:00	Autodesk Inventor Part	74 KB
Pin.ipt	02.03.2012 16:58	Autodesk Inventor Part	67 KB

Figure 4: Local file cache: After saving to Innovator



Name	Änderungsdatum	Typ	Größe
OldVersions	05.03.2012 11:46	Dateiordner	
iAssembly_with_Part.iam	05.03.2012 11:46	Autodesk Inventor Asse...	89 KB
Disc.ipt	02.03.2012 17:00	Autodesk Inventor Part	74 KB
Pin.ipt	02.03.2012 16:58	Autodesk Inventor Part	67 KB
Disc.ipt.CLB	05.03.2012 11:46	CLB-Datei	1 KB
iAssembly_with_Part.iam.CLB	05.03.2012 11:46	CLB-Datei	1 KB
Pin.ipt.CLB	05.03.2012 11:46	CLB-Datei	1 KB
lockfile.lck	05.03.2012 11:54	LCK-Datei	16 KB

Figure 5: Inventor: iAssembly (variant 1)

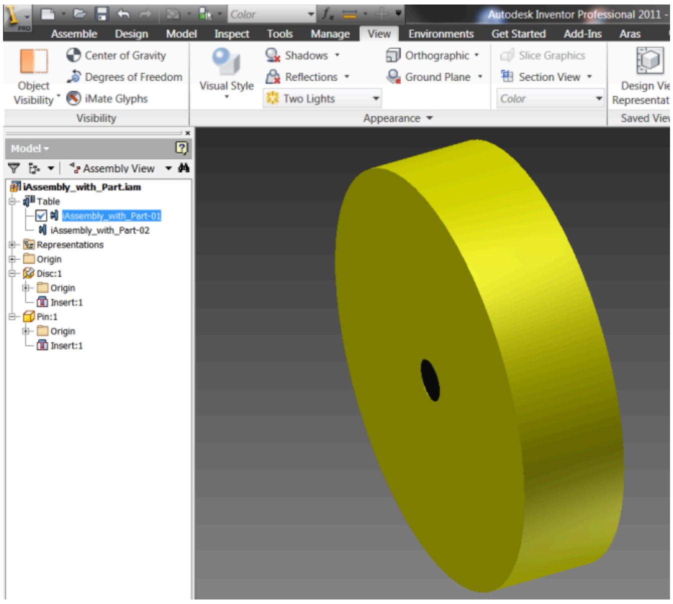


Figure 6: iAssembly (variant 2)

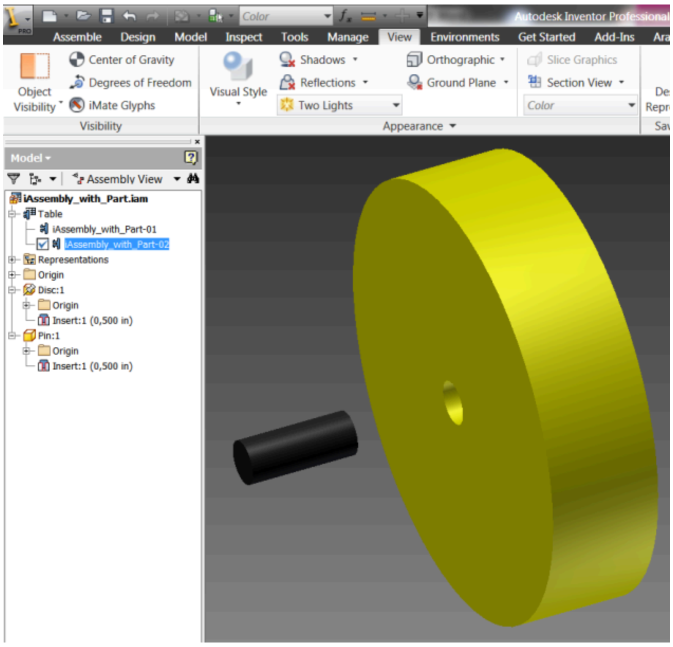
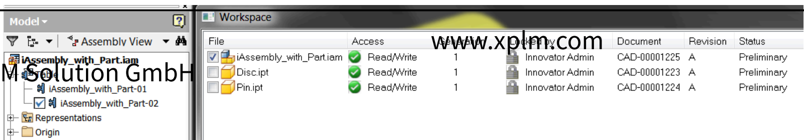


Figure 7: Inventor: Workspace Browser: iAssembly saved to Innovator (3 CAD documents created)

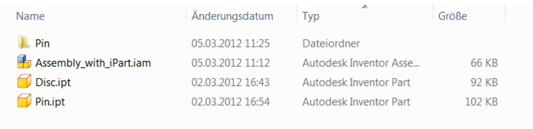


Only one CAD document is managed for the variant assembly file. It is linked to two part items for the management of the variance.

3.10.4.2 Assembly with iPart (component variance)

Example:

Figure 12: Local file cache: Before saving to Innovator (1 assembly, 1 part, 1 iPart)

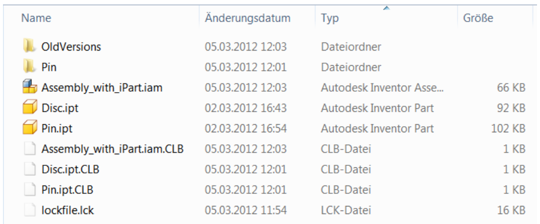


Name	Änderungsdatum	Typ	Größe
Pin	05.03.2012 11:25	Dateiordner	
Assembly_with_iPart.iam	05.03.2012 11:12	Autodesk Inventor Asse...	66 KB
Disc.ipt	02.03.2012 16:43	Autodesk Inventor Part	92 KB
Pin.ipt	02.03.2012 16:54	Autodesk Inventor Part	102 KB

Figure 13: Local file cache: Before saving to Innovator (2 variants of the iPart)

Name	Änderungsdatum	Typ	Größe
Pin-01.ipt	02.03.2012 16:54	Autodesk Inventor Part	74 KB
Pin-02.ipt	02.03.2012 16:56	Autodesk Inventor Part	72 KB

Figure 14: Local file cache: After saving to Innovator

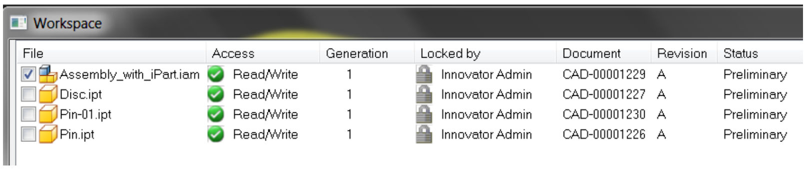


Name	Änderungsdatum	Typ	Größe
OldVersions	05.03.2012 12:03	Dateiordner	
Pin	05.03.2012 12:01	Dateiordner	
Assembly_with_iPart.iam	05.03.2012 12:03	Autodesk Inventor Asse...	66 KB
Disc.ipt	02.03.2012 16:43	Autodesk Inventor Part	92 KB
Pin.ipt	02.03.2012 16:54	Autodesk Inventor Part	102 KB
Assembly_with_iPart.iam.CLB	05.03.2012 12:03	CLB-Datei	1 KB
Disc.ipt.CLB	05.03.2012 12:01	CLB-Datei	1 KB
Pin.ipt.CLB	05.03.2012 12:01	CLB-Datei	1 KB
lockfile.lck	05.03.2012 11:54	LCK-Datei	16 KB

Figure 15: Local file cache: After saving to Innovator

Name	Änderungsdatum	Typ	Größe
lockfile.lck	05.03.2012 11:54	LCK-Datei	16 KB
Pin-01.ipt	02.03.2012 16:54	Autodesk Inventor Part	74 KB
Pin-01.ipt.CLB	05.03.2012 12:02	CLB-Datei	1 KB
Pin-02.ipt	02.03.2012 16:56	Autodesk Inventor Part	72 KB
Pin-02.ipt.CLB	05.03.2012 12:01	CLB-Datei	1 KB

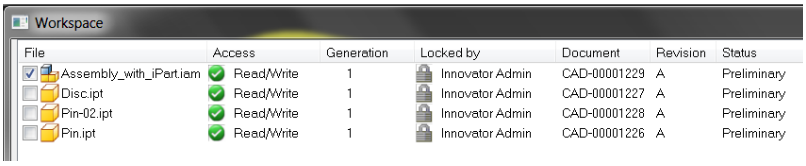
Figure 16: Inventor: Workspace Browser: Assembly with iPart variant 1 saved to Innovator



Workspace

File	Access	Generation	Locked by	Document	Revision	Status
Assembly_with_iPart.iam	Read/Write	1	Innovator Admin	CAD-00001229	A	Preliminary
Disc.ipt	Read/Write	1	Innovator Admin	CAD-00001227	A	Preliminary
Pin-01.ipt	Read/Write	1	Innovator Admin	CAD-00001230	A	Preliminary
Pin.ipt	Read/Write	1	Innovator Admin	CAD-00001226	A	Preliminary

Figure 17: Inventor: Workspace Browser: Assembly with iPart variant 2 saved to Innovator



Workspace

File	Access	Generation	Locked by	Document	Revision	Status
Assembly_with_iPart.iam	Read/Write	1	Innovator Admin	CAD-00001229	A	Preliminary
Disc.ipt	Read/Write	1	Innovator Admin	CAD-00001227	A	Preliminary
Pin-02.ipt	Read/Write	1	Innovator Admin	CAD-00001228	A	Preliminary
Pin.ipt	Read/Write	1	Innovator Admin	CAD-00001226	A	Preliminary

Figure 18: Inventor: iPart variant 1

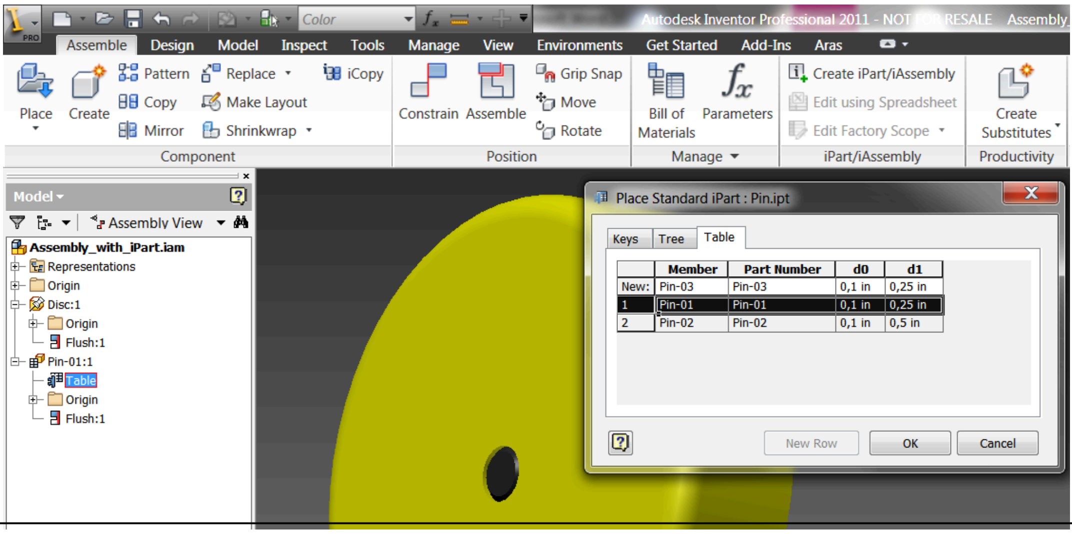
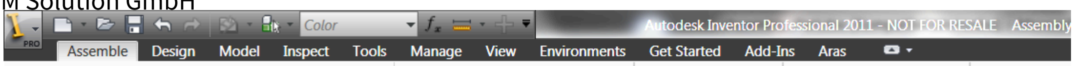


Figure 19: Inventor: iPart variant 2



Three CAD documents are managed for the one iPart which is used in two different variants.

- CAD Doc 1 for the generic (without part item)
- CAD Doc 2 for the first instance (with part item)
- CAD Doc 3 for the second instance (with part item)

3.10.4.3 iAssembly with iPart (assembly and component variance)

Example:

Figure 23: Local file cache: Before saving to Innovator (1 iAssembly, 1 part, 1 iPart)

Name	Änderungsdatum	Typ	Größe
Pin	20.03.2012 15:41	Dateiordner	
iAssembly_with_iPart.iam	20.03.2012 15:39	Autodesk Inventor Asse...	89 KB
Disc.ipt	02.03.2012 17:00	Autodesk Inventor Part	74 KB
Pin.ipt	20.03.2012 15:22	Autodesk Inventor Part	93 KB

Figure 24: Local file cache: Before saving to Innovator (2 variants of the iPart)

Name	Änderungsdatum	Typ	Größe
Pin-01.ipt	20.03.2012 15:22	Autodesk Inventor Part	69 KB
Pin-02.ipt	20.03.2012 15:23	Autodesk Inventor Part	67 KB

Figure 25: Local file cache: After saving to Innovator

Name	Änderungsdatum	Typ	Größe
OldVersions	20.03.2012 16:35	Dateiordner	
Pin	20.03.2012 16:35	Dateiordner	
iAssembly_with_iPart.iam	20.03.2012 16:35	Autodesk Inventor Assembly	89 KB
Disc.ipt	02.03.2012 17:00	Autodesk Inventor Part	74 KB
Pin.ipt	20.03.2012 15:22	Autodesk Inventor Part	93 KB
Disc.ipt.CLB	20.03.2012 16:33	CLB-Datei	1 KB
iAssembly_with_iPart.iam.CLB	20.03.2012 16:35	CLB-Datei	1 KB
Pin.ipt.CLB	20.03.2012 16:33	CLB-Datei	1 KB
lockfile.lck	20.03.2012 16:31	LCK-Datei	16 KB

Figure 26: Local file cache: After saving to Innovator

Name	Änderungsdatum	Typ	Größe
Pin-01.ipt	20.03.2012 15:22	Autodesk Inventor Part	69 KB
Pin-02.ipt	20.03.2012 15:23	Autodesk Inventor Part	67 KB
Pin-01.ipt.CLB	20.03.2012 16:33	CLB-Datei	1 KB
Pin-02.ipt.CLB	20.03.2012 16:35	CLB-Datei	1 KB
lockfile.lck	20.03.2012 15:41	LCK-Datei	16 KB

Figure 27: CAD: iAssembly variant 1 and iPart variant 2

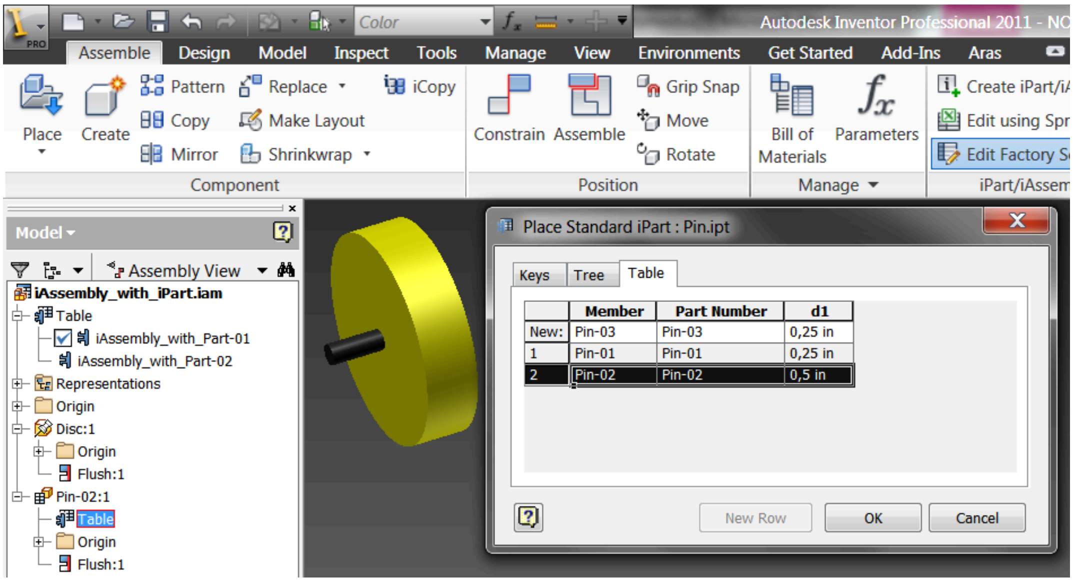
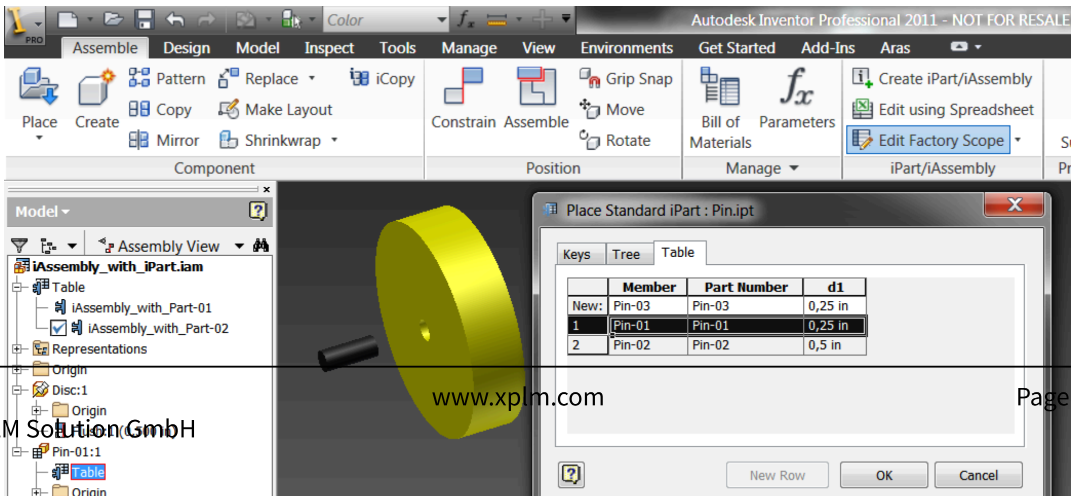


Figure 28: Inventor: iAssembly variant 2 and iPart variant 1



- One CAD document is managed for the variant assembly file. It is linked to three part items for the management of the variance.
- Three CAD documents are managed for the one iPart which is used in 2 different variants.
  - CAD Doc 1 for the generic (without part item)
  - CAD Doc 2 for the first instance (with part item)
  - CAD Doc 3 for the second instance (with part item)